11. Adaptive Selection and Staged Continuation

11.1 Cumulative phase construction for adaptive selection

If a shortlisted macro generator m admits a split family $\mathcal{C}_{\text{split}}(m) = \{m_1, \ldots, m_{K_m}\}$, then each split child inherits the same position p and therefore defines a child record $r_j = (m_j, p)$. For simplicity we drop the j, and assume each m below is a unique generator.

Record-first view: the primary object is a candidate-position record $r := (m, p)$, with projections $\pi_m(m, p) := m$ and $\pi_p(m, p) := p$. For each phase $k \in \{1, 2, 3\}$, the one-way selector flow is

$$\mathcal{R}_k \longrightarrow \mathcal{S}_k \longrightarrow r_k^\star,$$

with

$$\mathcal{R}_k := \{(m, p) : m \in \mathcal{G}_k,\ p \in \mathcal{P}_m^k\}, \qquad \mathcal{S}_k := \{(m, p) \in \mathcal{R}_k : S_k(m, p) \geq \tau_k\} \quad \text{or} \quad \mathrm{Top}_{N_k}(S_k; \mathcal{R}_k)\,, \qquad r_k^\star \in \mathcal{S}_k.$$

where S_k denotes the phase-k score used for shortlist construction. Generator image and position image: $\mathcal{G}^{(k)} := \pi_m(\mathcal{S}_k)$ and $\mathcal{P}_m^{(k)} := \{p : (m, p) \in \mathcal{S}_k\}$. Record inheritance: when later phases inherit the same candidate-position records, the clean handoff is $\mathcal{R}_{k+1} := \mathcal{S}_k$. Hence the cumulative record chain is $\mathcal{R}_3 \subseteq \mathcal{R}_2 \subseteq \mathcal{R}_1$, the cumulative generator chain is $\mathcal{G}^{(3)} \subseteq \mathcal{G}^{(2)} \subseteq \mathcal{G}^{(1)} \subseteq \mathcal{G}$, the first chain follows from $\mathcal{R}_{k+1} := \mathcal{S}_k \subseteq \mathcal{R}_k$, and the retained position images inherit analogously via the corresponding record shortlists $\mathcal{S}_1 \to \mathcal{S}_2 \to \mathcal{S}_3$. Post-shortlist admission, batching, pruning, and beam continuation are then defined in terms of the final Phase-3 surface.

Seed broadening: in addition, we use a smooth broadening of the initial candidate record $\mathcal{R}$ as expected generator utility saturates, anti-comensurate to scaffold depth, where the initial seeds are low-order generators and late-stage seeds are higher-order excitation and coupling terms.

Chronology: thus the presentation chronologically accords with the algorithmic implementation: Phase 1 $\to$ Phase 2 $\to$ Phase 3, with each to-arrow filtering its preceding phase.

11.2 Phase 1

11.2.1 Core signal, append position, and one-coordinate local model

Core record, scaffold, and append slot: $r = (m, p)$, $\mathcal{O} = (a_1, \ldots, a_n)$, $m \in \mathcal{G}$, and $p \in \mathcal{P}_m^1 = \{0, \ldots, n\}$. Here m is a candidate generator already admitted by the generator gate (so $m \in \mathcal{G}_1$), and p is the index of a slot in the pre-insertion, ordered scaffold $\mathcal{O}$. Window bias: note, the insertion window near the terminal scaffold point is biased to be small proportional to expected scaffold utility remaining.

Scaffold parameter space and energy map: WLOG consider $\mathcal{O}$, s.t. each a is assigned one scalar $\theta(\mod 2\pi)$, then define $\Theta_{\mathcal{O}} := \mathbb{R}^{|\mathcal{O}|}$, $E(\mathcal{O}, \theta) := \langle \psi(\mathcal{O}, \theta) | H | \psi(\mathcal{O}, \theta) \rangle$, and $\theta \in \Theta_{\mathcal{O}}$. Scaffold insertion and parameter insertion: define $\mathcal{O} \oplus_p m := (a_1, \ldots, a_p, m, a_{p+1}, \ldots, a_n)$ and $\theta \oplus_p \vartheta := (\theta_1, \ldots, \theta_p, \vartheta, \theta_{p+1}, \ldots, \theta_n) \in \Theta_{\mathcal{O} \oplus_p m}$.

Write $|\psi\rangle := U_{>p} U_{\leq p} |\psi_0\rangle$, $\widetilde{A}_r := U_{>p} A_m U_{>p}^\dagger$, $U_r(\alpha) := e^{-i\alpha \widetilde{A}_r}$, and $|\psi_r(\alpha)\rangle := U_r(\alpha)|\psi\rangle$. Let $Q_\psi := I - |\psi\rangle\langle\psi|$, $|d_r\rangle := \partial_\alpha |\psi_r(\alpha)\rangle|_{\alpha=0} = -i\widetilde{A}_r|\psi\rangle$, $\tau_r := Q_\psi |d_r\rangle$, $g_r := 2\Re\langle\psi|H|d_r\rangle$, $\sigma_r := \mathrm{SE}[\widehat{g}(r)] \geq 0$, $g_{\text{lcb}}(r) := \max\{|g_r| - z_\alpha \sigma_r, 0\}$, and $F_{\text{raw}}(r) := \langle \tau_r, \tau_r \rangle + \varepsilon_F$. The live broad Phase-1 one-coordinate trust-region proxy is

$$\Delta E_{1,\text{TR}}(r) := \max_{|\alpha| \leq \rho/\sqrt{F_{\text{raw}}(r)}} \left[g_{\text{lcb}}(r)|\alpha| - \frac{1}{2}\lambda_F F_{\text{raw}}(r)\alpha^2 \right].$$

The strict one-coordinate insertion drop is retained as the exact auxiliary comparator

$$\delta E_1(m,p \mid \mathcal{O},\theta) = E(\mathcal{O},\theta) - \inf_{\vartheta\in\mathbb{R}} E(\mathcal{O}\oplus_p m, \theta\oplus_p \vartheta).$$

The infimum notation may be replaced by min or Min whenever the minimizer exists. Thus $\Delta E_{1,\mathrm{TR}}(r)$ is the live broad-pool second-order local insertion proxy, whereas $\delta E_1(m,p \mid \mathcal{O},\theta)$ is the exact one-coordinate insertion drop. [1]

11.2.2 Gates, burdens, and feature ingredients

Base hardware ingredients: each candidate-position pair carries the proxy triple $C_{2\mathrm{q}}(m,p)$, $D(m,p)$, and $\Delta K(m,p)$, with fixed nonnegative weights $\lambda_{2\mathrm{q}}, \lambda_d, \lambda_\theta$.[2] Here $C_{2\mathrm{q}}$ is a two-qubit proxy cost, D is an expected depth-induced by m (at p) proxy, and ΔK is the marginal parameter count. Positivity assumptions: assume throughout that $\lambda_{2\mathrm{q}} \geq 0$, $C_{2\mathrm{q}}(m,p) \geq 0$, and $\lambda_\theta, \lambda_d, \Delta K(m,p), D(m,p) > 0$ on admissible records. [3]

11.2.3 Anticipated utility left and controller schedule

Let t denote the current selector step, let $D_{\mathrm{left}}(t)$ be the remaining controller depth budget, and let $\widehat{N}_{\mathrm{rem}}(t)$ be a heuristic forecast (expected count) of useful remaining admissions. For gain-process statistics, define $\Delta_{\mathrm{adm}}(s) := E(\theta^{(s-1)}) - E(\theta^{(s)})$, $\Delta^+_{\mathrm{adm}}(s) := \operatorname{asinh}\Big(\frac{\Delta_{\mathrm{adm}}(s)}{\tau_\Delta}\Big)$, and let $\mathcal{I}^+_t$ be the index set of past selector steps with valid admissions used for EWMA/MAD statistics.

Let D_{allow} denote the total allowed depth budget, let $\ell^{\mathrm{plat}}_t := \max\{1, \lceil P\, D_{\mathrm{left}}(t)/D_{\mathrm{allow}}\rceil\}$ and $\ell^{\mathrm{low}}_t := \max\{1, \lceil L\, D_{\mathrm{left}}(t)/D_{\mathrm{allow}}\rceil\}$, and define the relative-flattening and absolute weak-gain thresholds by $\tau^+_{\mathrm{plat}}(t) := (\eta_0 + \eta_1(1 - D_{\mathrm{left}}(t)/D_{\mathrm{allow}}))m_t$ and $\tau^+_{\mathrm{low}}(t) := \operatorname{asinh}(\tau_{\mathrm{use},t}/\tau_\Delta)$. The corresponding depth-adaptive streak statistics are $\mathrm{plateau_streak}(t) := \frac{P}{\ell^{\mathrm{plat}}_t}\max\{q \in \{0,\dots,\ell^{\mathrm{plat}}_t\} : \Delta^+_{\mathrm{adm}}(t-j+1) \leq \tau^+_{\mathrm{plat}}(t)\ \forall 1 \leq j \leq q\}$ and $\mathrm{low_streak}(t) := \frac{L}{\ell^{\mathrm{low}}_t}\max\{q \in \{0,\dots,\ell^{\mathrm{low}}_t\} : \Delta^+_{\mathrm{adm}}(t-j+1) \leq \tau^+_{\mathrm{low}}(t)\ \forall 1 \leq j \leq q\}$.

For stagnation, gain, decay, and scale summaries, define $u_{\mathrm{stag}}(t) := \mathrm{clip}\Big(\max\Big(\frac{\mathrm{plateau_streak}(t)}{P}, \frac{\mathrm{low_streak}(t)}{L}\Big), 0, 1\Big)$, $m_t := \mathrm{EWMA}_{s\in\mathcal{I}^+_t}(\Delta^+_{\mathrm{adm}}(s))$, $\rho_t := \mathrm{clip}\Big(\frac{\mathrm{EWMA}_{\mathrm{recent}}(\Delta_{\mathrm{adm}})}{\mathrm{EWMA}_{\mathrm{older}}(\Delta_{\mathrm{adm}})+\varepsilon}, \rho_{\min}, 1\Big)$, $\gamma_t := [-\log\rho_t]_+$, and $s_t := \max\Big(s_{\min}, \mathrm{MAD}_{s\in\mathcal{I}^+_t}(\Delta^+_{\mathrm{adm}}(s))\Big)$.

For the frontier statistic, let $s^{(1)}_{\mathrm{raw},1}(t) \geq s^{(1)}_{\mathrm{raw},2}(t) \geq 0$ denote the first two order statistics of the current broad Phase-1 score multiset $\{S_1(r;t) : r \in \mathcal{R}_1\}$ before any $\tau_1(t)$-thresholding or $N_1(t)$-truncation, with the conventions $s^{(1)}_{\mathrm{raw},1}(t) = s^{(1)}_{\mathrm{raw},2}(t) = 0$ if $\mathcal{R}_1 = \varnothing$ and $s^{(1)}_{\mathrm{raw},2}(t) = 0$ if $|\mathcal{R}_1| = 1$. Define

$$u_{\mathrm{front}}(t) := \frac{s^{(1)}_{\mathrm{raw},2}(t) + \varepsilon}{s^{(1)}_{\mathrm{raw},1}(t) + \varepsilon}.$$

[1] User interest: How does window parameterization width scale with cost?

[2] Admissibility gate: For each selector stage k, let $\Gamma_k : \mathcal{G}^{\mathrm{in}}_k \to \{0,1\}$ be the generator-level admissibility gate and define $\mathcal{G}^{\mathrm{adm}}_k := \{m \in \mathcal{G}^{\mathrm{in}}_k : \Gamma_k(m) = 1\}$. Record domain: the working candidate-record domain is built only from this gated generator pool, $\mathcal{R}_k := \{(m,p) : m \in \mathcal{G}^{\mathrm{adm}}_k,\ p \in \mathcal{P}^{\mathrm{in},k}_m\}$. Selector action: all selectors below act on $\mathcal{R}_k$ itself, so admissibility is enforced by pre-score domain construction rather than post hoc score masking. Tie order: fix once and for all a total order $\prec$ on records (for example, lexicographic order on (m,p)). Tie-broken maximizer: for any score $S : \mathcal{R}_k \to \mathbb{R}$, define $\arg\max^{\prec}_{r\in\mathcal{R}_k} S(r) := \min_{\prec}\{r \in \mathcal{R}_k : S(r) = \max_{u\in\mathcal{R}_k} S(u)\}$, and similarly for $\arg\min^{\prec}$. Shortlist surface: define $\mathrm{Top}^{\prec}_K(S;\mathcal{R}_k)$ as the ordered list of the first K records obtained by sorting $\mathcal{R}_k$ in descending score, with ties broken by $\prec$. Unadorned arg max, arg min, and Top_K below follow this convention.

[3] Depth and parameter count are constants for our L=2 pareto lean runs.

Thus $u_{\text{front}}(t) \in [0,1]$, with larger values corresponding to a flatter, more crowded current frontier and smaller values corresponding to a clearly separated leading record.[4]

Then the survival hazard is $h_t(j) := \sigma\Big(\beta_0 + \beta_{\text{stag}} u_{\text{stag}}(t) + \beta_{\text{front}} u_{\text{front}}(t) + \eta_h(j-1)\Big)$.

$$S_t(k) := \prod_{j=1}^{k-1} (1 - h_t(j)), \qquad Q_t(k) := \Phi\left(\frac{m_t - \gamma_t(k-1) - \operatorname{asinh}\Big(\frac{\tau_{\text{use},t}}{\tau_\Delta}\Big)}{s_t}\right).$$

$$\widehat{N}_{\text{rem}}(t) := \sum_{k=1}^{D_{\text{left}}(t)} S_t(k)\, Q_t(k).$$

For pessimistic/optimistic runway envelopes, set $h_t^{\text{pess/opt}}(j) := \sigma\big(\operatorname{logit} h_t(j) \pm \Delta_h\big)$, $m_t^{\text{pess/opt}} := m_t \mp \Delta_m$, $s_t^{\text{pess/opt}} := s_t + \Delta_s$ (pess) or $s_t^{\text{pess/opt}} := \max(s_{\min}, s_t - \Delta_s)$ (opt), $S_t^{\text{pess/opt}}(k) := \prod_{j=1}^{k-1}(1 - h_t^{\text{pess/opt}}(j))$, and $Q_t^{\text{pess/opt}}(k) := \Phi\left(\frac{m_t^{\text{pess/opt}} - \gamma_t(k-1) - \operatorname{asinh}\Big(\frac{\tau_{\text{use},t}}{\tau_\Delta}\Big)}{s_t^{\text{pess/opt}}}\right)$. Then $\widehat{N}_{\text{rem}}^{\text{low/high}}(t) := \sum_{k=1}^{D_{\text{left}}(t)} S_t^{\text{pess/opt}}(k)\, Q_t^{\text{pess/opt}}(k)$, with $(\text{low}, \text{high}) \leftrightarrow (\text{pess}, \text{opt})$, and the confidence ratio is $C_t := \frac{\widehat{N}_{\text{rem}}^{\text{low}}(t)}{\widehat{N}_{\text{rem}}^{\text{high}}(t) + \varepsilon}$.

For the controller coordinate and directional transforms, define

$$R_t := \frac{\widehat{N}_{\text{rem}}(t)}{D_{\text{left}}(t) + \varepsilon} \in [0,1], \qquad e_t := R_t^a, \qquad g_t := (1 - R_t)^b, \qquad a, b > 0.$$

Here R_t is remaining useful runway as a fraction of the depth-budget permitted by hardware/runtime constraints, and g_t is the late-stage activation coordinate that scales controller variables depending on age, where age is a normalized quantity relative to/determined by expected utility of more generators.

Then $R_t \approx 1$ early and $R_t \approx 0$ late. Use e_t for early-dominant quantities and g_t for late-dominant quantities. In particular, define

$$\tau_{\text{pos}}(t) := \tau_{\text{pos}}^{\min} + (\tau_{\text{pos}}^{\max} - \tau_{\text{pos}}^{\min}) e_t, \qquad \lambda_F(t) := \lambda_F^{\min} + (\lambda_F^{\max} - \lambda_F^{\min}) g_t, \qquad u_{\text{sat}}(t) := 1 - R_t.$$

The quantity $\widehat{N}_{\text{rem}}(t)$ is controller telemetry rather than a physical observable: it summarizes recent selector progress and remaining useful runway, but it is not itself a hard admissibility certificate.[5]

For inherited refit windows, use the maturity-shrinking retention law

$$\rho_W(t) := \rho_W^{\min} + (\rho_W^{\max} - \rho_W^{\min}) e_t, \qquad 0 \le \rho_W^{\min} \le \rho_W^{\max} \le 1,$$

so that the retained old-coordinate fraction is largest early and contracts as useful runway is exhausted.

[4] With the empty-pool convention $s_{\text{raw},1}^{(1)}(t) = s_{\text{raw},2}^{(1)}(t) = 0$, one gets $u_{\text{front}}(t) = 1$. This treats the absence of a usable current frontier as maximal compression rather than as a missing-data null signal. An explicit signal-to-noise controller statistic is a possible implementation here, but it may not be necessary if worsening signal-to-noise already manifests as frontier flattening, which the present expected-utility model scores directly through $u_{\text{front}}(t)$. Here $h_t(j)$ is the conditional probability that usefulness terminates at future offset j, given survival through offset $j-1$; accordingly, $S_t(k)$ is the survival probability to the start of offset k.

[5] Interpretation: the model has three layers — (i) gain process (m_t, s_t), (ii) horizon-decay process (γ_t), and (iii) survival process $(h_t(j))$; together these drive the runway forecast $\widehat{N}_{\text{rem}}(t)$.

For measurement effort, use the Phase-1 baseline schedule

$$N_{\text{shot},1}(t) := \Big\lceil N_{\text{shot}}^{\min} + (N_{\text{shot}}^{\max} - N_{\text{shot}}^{\min})\Big(\alpha_{\text{front}}\, u_{\text{front}}(t) + \alpha_{\text{util}}\,(1 - R_t)\Big)\Big\rceil,$$

with $N_{\text{shot}}^{\min} \le N_{\text{shot}}^{\max}$, $\alpha_{\text{front}}, \alpha_{\text{util}} \ge 0$, and $\alpha_{\text{front}} + \alpha_{\text{util}} = 1$. Thus Phase 1 increases shot count when the frontier flattens and when useful runway is being exhausted.

For $k \in \{2,3\}$, once the phase-k uncertainty summary $u_{\sigma,k}(t)$ is available, use

$$N_{\text{shot},k}(t) := \Big\lceil N_{\text{shot}}^{\min} + (N_{\text{shot}}^{\max} - N_{\text{shot}}^{\min})\Big(\alpha_{\text{front}}\, u_{\text{front}}(t) + \alpha_{\text{util}}\,(1 - R_t) + \alpha_{\sigma}\, u_{\sigma,k}(t)\Big)\Big\rceil, \qquad k \in \{2,3\},$$

with $\alpha_{\text{front}}, \alpha_{\text{util}}, \alpha_{\sigma} \ge 0$ and $\alpha_{\text{front}} + \alpha_{\text{util}} + \alpha_{\sigma} = 1$.[6] Thus Phase 2 and Phase 3 inherit the same controller baseline and then increase shot count further when geometric uncertainty rises.

For phase thresholds, use the live monotone law

$$\tau_k(t) := \tau_k^{\min} + (\tau_k^{\max} - \tau_k^{\min})e_t, \qquad \tau_k^{\min} \le \tau_k^{\max}, \qquad k \in \{1,2,3\},$$

so that shortlist admission is stricter early and more permissive later.

For phase shortlist caps, use the live bounded law

$$N_k(t) := \Big\lceil N_k^{\min} + (\,N_k^{\max} - N_k^{\min}\,)g_t\Big\rceil, \qquad N_k^{\min} \le N_k^{\max}, \qquad k \in \{1,2,3\}.$$

Thus $N_k(t)$ and $\tau_k(t)$ are both live shortlist-control quantities: $N_k(t)$ ensures budget constraints are met, whereas $\tau_k(t)$ helps ensure logical admissibility constraints proceed with time.

11.2.4 Phase-local cost and structural burden definitions

For any current phase-input record family $\mathcal{R}_k(t)$, with $\mathcal{R}_1(t) := \mathcal{R}_1$ and $k \in \{1,2,3\}$, and any scalar record feature $x : \mathcal{R}_k(t) \to \mathbb{R}_{\ge 0}$, define the empirical quantiles $q_\alpha^x(t) := \text{Quantile}_\alpha(\{x(u) : u \in \mathcal{R}_k(t)\})$ and $m_x(t) := q_{0.5}^x(t)$, together with the robust scales $s_x^{\text{MAD}}(t) := 1.4826\ \text{median}_{u\in\mathcal{R}_k(t)}|x(u) - m_x(t)|$, $s_x^{\text{pct}}(t) := \frac{q_{0.9}^x(t) - q_{0.1}^x(t)}{2.563}$, and $s_x(t) := \max(\varepsilon_x,\ s_x^{\text{MAD}}(t),\ s_x^{\text{pct}}(t))$. Thus $s_x(t)$ is a percentile/MAD hybrid reference scale, with percentile fallback when the MAD degenerates on sparse discrete features.

Define the one-sided robust normalized excess by $z_x(r;t) := \text{asinh}\Big(\frac{[x(r) - m_x(t)]_+}{s_x(t)}\Big)$, with $[y]_+ := \max(y, 0)$. When a direct ratio form is preferred, use the reference level $x_{\text{ref}}(t) := m_x(t) + s_x(t)$.

For a candidate-position record $r = (m, p)$, let $\mathcal{L}_r := \text{supp}(m) = \{j : m_j \ne I\}$, $\text{wt}(r) := |\mathcal{L}_r|$, and $\#_{X,Y}(r) := |\{j \in \mathcal{L}_r : m_j \in \{X, Y\}\}|$.

Define feature-specific robust excesses by $z_{\text{wt}}(r;t) := z_x(r;t)$ with $x = \text{wt}$ and $z_{XY}(r;t) := z_x(r;t)$ with $x = \#_{X,Y}$. The structural burden term is

$$D_{\text{struct}}(r;t) := \beta_{\text{wt}}\, z_{\text{wt}}(r;t) + \beta_{XY}\, z_{XY}(r;t),$$

with nonnegative coefficients $\beta_{\text{wt}}, \beta_{XY} \ge 0$.[7]

For candidate-specific measurement burden, let $O_r := i[H, \widetilde{A}_r] = \sum_{b\in\mathcal{B}_r} O_{r,b}$, with $O_{r,b} := \sum_{\ell\in B_{r,b}} c_{r,b\ell} P_{r,b\ell}$, where $\mathcal{B}_r$ is the qubit-wise commuting partition of the candidate's gradient

[6] A one-weight simplification is the special case $\alpha_{\text{front}} = \alpha_{\text{util}} = \frac{1-\alpha_\sigma}{2}$.

[7] Provenance note: generator novelty $G_{\text{new}}(r)$, family novelty $\text{groups}_{\text{new}}(r)$ with single-valued $\mathfrak{g} : \mathcal{G} \to \mathcal{C}$, lifetime/staleness $\mathbf{1}_{\text{lifetime}}(r;t)$ with $t_{\text{first}}(m)$ meaning first time ever seen in the run, the auxiliary raw burden $D_{\text{raw}}(r;t) := D_{\text{struct}}(r;t) + d_{\text{pos}}(r)$ with $d_{\text{pos}}(r) := |p - p_{\text{app}}|$, and the normalized descendants $\bar{D}_{\text{life}}^{\text{struct}}, \bar{G}_{\text{new}}, \bar{C}_{\text{new}}, \bar{c}, \bar{P}_{\text{opt}}$ are omitted from the live hardware cost in the present draft and retained only as controller-side provenance/policy bookkeeping.

observable and $\mathcal{B}_r^{\text{new}} \subseteq \mathcal{B}_r$ is the subset not already covered by measurement reuse. Here $\widetilde{A}_r$ is the dressed candidate generator at record $r = (m, p)$, $\mathcal{B}_r$ is the partition of $O_r = i[H, \widetilde{A}_r]$ into qubit-wise commuting Pauli-string measurement groups, and $\mathcal{B}_r^{\text{new}}$ is the subset that still requires new measurement effort. When exact state-dependent variances are unavailable, use the commutator-based proxy

$$\widehat{v}_{r,b} := \sum_{\ell \in B_{r,b}} |c_{r,b\ell}|^2, \qquad \widehat{\widehat{C}}_{\text{shot}}(r) := \frac{1}{\text{shot}_{\text{ref}}} \cdot \frac{\left(\sum_{b \in \mathcal{B}_r^{\text{new}}} \sqrt{\widehat{v}_{r,b}}\right)^2}{\sigma_\star^2}.$$

Here $\sigma_\star$ is the target standard error for the candidate gradient estimate. The positive constant shot_{ref} is a baseline measurement budget against which candidate-specific shot cost is compared.

For the shared phase-k cost, define $z_{2\text{q}}(r;t) := z_x(r;t)$ with $x = C_{2\text{q}}$, $z_d(r;t) := z_x(r;t)$ with $x = D$, $z_\theta(r;t) := z_x(r;t)$ with $x = \Delta K$, $z_{\text{shot}}(r;t) := z_x(r;t)$ with $x = \widehat{\widehat{C}}_{\text{shot}}$, and

$$K_k(r;t) := \lambda_{2\text{q}}\, z_{2\text{q}}(r;t) + \lambda_d\, z_d(r;t) + \lambda_\theta\, z_\theta(r;t) + \lambda_{\text{shot}}\, z_{\text{shot}}(r;t) + D_{\text{struct}}(r;t).$$

Explicitly,

$$\begin{aligned} K_k(r;t) = \lambda_{2\text{q}}\, z_{2\text{q}}(r;t) + \lambda_d\, z_d(r;t) + \lambda_\theta\, z_\theta(r;t) + \lambda_{\text{shot}}\, z_{\text{shot}}(r;t) \\ + \beta_{\text{wt}}\, z_{\text{wt}}(r;t) + \beta_{XY}\, z_{XY}(r;t), \qquad k \in \{1,2,3\}. \end{aligned}$$

11.2.5 Final Phase I Scoring

For Phase 1, the set $\mathcal{R}_1 := \{(m,p) : m \in \mathcal{G}_1,\ p \in \mathcal{P}_m^1\}$ is scored by

$$S_1(r;t) := \frac{\Delta E_{1,\text{TR}}(r)}{1 + K_1(r;t)}, \qquad r \in \mathcal{R}_1.$$

Here $\Delta E_{1,\text{TR}}(r)$ is the Phase-1 one-coordinate trust-region proxy defined in §11.2.1, and the phase-1 cost is evaluated on the current Phase-1 input family. This Phase-1 score is intentionally broad as a screening surface: it privileges cheap local second-order insertion estimates while the controller still has long useful runway. The corresponding broad Phase-1 probing/evaluation budget is set by the baseline controller shot law $N_{\text{shot},1}(t)$.

In the present draft, the live Phase-1 numerator is the one-coordinate trust-region proxy $\Delta E_{1,\text{TR}}(r)$, so that the broad screen remains in the same gradient-plus-second-order family later refined in Phases 2 and 3. The strict one-coordinate insertion drop $\delta E_1(r \mid \mathcal{O}, \theta)$ is retained only as an auxiliary exact comparator for selective frontier confirmation, ablation studies, or future confirm-stage refinement. A broader early-stage local-refit numerator remains only a non-live hook. To make such a refinement live, one would first have to specify (i) the admissible refit-window family, (ii) the corresponding constrained local-refit drop, and (iii) a controller activation law independent of the current raw Phase-1 score multiset, so that the live Phase-1 law is not self-referential.

11.2.6 Position probing, trough detection, and effective selector

The Phase-1 scoring mechanism produces the retained record shortlist for further refined scoring. The retained model set is $\mathcal{G}^{(1)} := \pi_m(\mathcal{S}_1(t))$, the retained position fibers are $\mathcal{P}_m^{(1)} := \{p : (m,p) \in \mathcal{S}_1(t)\}\ \forall m \in \mathcal{G}^{(1)}$, the Phase-2 record pool is $\mathcal{R}_2 := \mathcal{S}_1(t)$, and the acceptance set is $\mathcal{A}_1(t) := \{r = (m,p) \in \mathcal{R}_1 : S_1(r;t) \geq \tau_1(t)\} = \{r_{(1)}, \ldots, r_{(M)}\}$, with $M := |\mathcal{A}_1(t)|$. The ordered Phase-1 scores

are $s_k^{(1)}(t) := S_1(r_{(k)}; t)$, with $s_1^{(1)}(t) \geq s_2^{(1)}(t) \geq \cdots \geq s_M^{(1)}(t)$, and the candidate score distinction is $u_{\text{front},k}^{(1)}(t) := \frac{s_{k+1}^{(1)}(t)+\varepsilon}{s_k^{(1)}(t)+\varepsilon}$, with $k = 1, \ldots, M-1$ and $0 < \eta_1 < 1$.

$$k_1^\star(t) := \begin{cases} 0, & M = 0, \\ \min\Big\{k \in \{1, \ldots, \min(M-1, N_1(t)-1)\} : u_{\text{front},k}^{(1)}(t) \leq \eta_1\Big\}, & \text{if this set is nonempty}, \\ \min\{M, N_1(t)\}, & \text{otherwise}, \end{cases}$$

$$\mathcal{D}_1(t) := \{r_{(j)}(t) : 1 \leq j \leq k_1^\star(t)\}, \qquad \mathcal{S}_1(t) := \mathcal{A}_1(t) \cap \mathcal{D}_1(t) = \mathcal{D}_1(t).$$

Here η_1 is a controller parameter chosen either from policy intuition or from empirical calibration.

11.3 Phase II

Phase II is the raw geometric selector stage. It acts on the Phase-1 survivors, restores the coarse geometric signal that distinguishes directional usefulness beyond insertion-drop alone, and produces the filtered shortlist $\mathcal{S}_2$.

11.3.1 Raw geometry, restored novelty, and auxiliary overlap window

The Phase-2 selector works over candidate-position records inherited from Phase 1. In the record-inheritance mode used here, the inherited and admissible domain is $\mathcal{R}_2 := \mathcal{S}_1(t)$. For $r = (m, p)$, let $W(r; t) \subseteq \{1, \ldots, n+1\}$ denote the inherited overlap window attached to r. In the present three-phase architecture this window enters the Phase-2 selector authoritatively only through overlap against the current horizontal tangent family; the same inherited window is then carried forward to Phase 3, where reduced-window elimination and final reranking are performed.[8]

Reusing the record-local insertion objects already introduced in Phase 1, let $|d_r^{(2)}\rangle := -\widetilde{A}_r^2|\psi\rangle$ and $\chi_r := \text{clip}\Big(1 - \frac{z_\alpha \sigma_r}{|g_r|+\varepsilon_g}, 0, 1\Big)$.[9] For phase-dependent shot allocation in later phases, define

$$u_{\sigma,k}(t) := \text{clip}\left(\text{median}_{r \in \mathcal{R}_k(t)} \frac{z_\alpha \, \sigma_r}{|g_r| + \varepsilon_g}, \, 0, \, 1\right), \qquad k \in \{2, 3\},$$

where in Phase 2 the record-local gradient and uncertainty estimates are evaluated using $N_{\text{shot},2}(t)$, and in Phase 3 the pair (g_r, σ_r) is inherited from the Phase-II raw geometry carried by the surviving records.

[8] Here p indexes the cut after U_p. Thus $U_{>p} := U_K \cdots U_{p+1}$ and $U_{\leq p} := U_p \cdots U_1$, and insertion at cut p means $|\psi_r(\alpha)\rangle = U_{>p} e^{-i\alpha A_m} U_{\leq p}|\psi_0\rangle = U_K \cdots U_{p+1} \, e^{-i\alpha A_m} \, U_p \cdots U_1|\psi_0\rangle$. So the new generator is inserted between U_p and U_{p+1}.

[9] Let $\widehat{g}(r)$ denote the empirical gradient estimator for record $r = (m, p)$ obtained from N_r independent shot-level repetitions. If $X_1^{(r)}, \ldots, X_{N_r}^{(r)}$ are the repetition-level gradient samples, then $\widehat{g}(r) := \frac{1}{N_r} \sum_{i=1}^{N_r} X_i^{(r)}$ and $\sigma_r := \text{SE}\big[\widehat{g}(r)\big] = \sqrt{\text{Var}\big(\widehat{g}(r)\big)}$. Under independent repetitions, $\text{Var}\big(\widehat{g}(r)\big) = \frac{\text{Var}\left(X_1^{(r)}\right)}{N_r}$, so in practice one estimates $\sigma_r \approx \frac{s_r}{\sqrt{N_r}}$, where s_r is the sample standard deviation of the repetition-level gradients. Thus σ_r is the standard error of the mean gradient estimate, not the sample standard deviation itself.

The raw geometric quantities are

$$
\begin{aligned}
F_{\mathrm{raw}}(r) &:= \langle \tau_r, \tau_r \rangle + \varepsilon_F, \qquad \varepsilon_F > 0, \\
\mathcal{N}_2(r;t) &:= 1 - \max_{j \in W(r;t)} \frac{\left|\langle d_r | Q_\psi | \partial_{\theta_j} \psi \rangle\right|}{\sqrt{F_{\mathrm{raw}}(r)}} \frac{1}{\sqrt{\langle \partial_{\theta_j} \psi | Q_\psi | \partial_{\theta_j} \psi \rangle + \varepsilon_F}}, \\
h_r &:= 2\Re\Big(\langle d_r | H | d_r \rangle + \langle \psi | H | d_r^{(2)} \rangle \Big), \\
\Delta E_{\mathrm{TR}}^{\mathrm{raw}}(r) &:= \max_{|\alpha| \le \rho/\sqrt{F_{\mathrm{raw}}(r)}} \left[g_{\mathrm{lcb}}(r)|\alpha| - \frac{1}{2} \max(h_r, 0)\alpha^2 \right].
\end{aligned}
$$

Here $F_{\mathrm{raw}}(r)$ is the raw tangent norm, $\mathcal{N}_2(r;t)$ is the restored Phase-2 novelty score, h_r is the raw local curvature, and $\Delta E_{\mathrm{TR}}^{\mathrm{raw}}(r)$ is the raw trust-region gain before reduced-window elimination.

The authoritative Phase-II score is

$$
S_2(r;t) := \frac{\Delta E_{\mathrm{TR}}^{\mathrm{raw}}(r)\, \chi_r\, \mathcal{N}_2(r;t)}{1 + K_2(r;t)}, \qquad r \in \mathcal{R}_2.
$$

Thus Phase II remains the raw geometric selector stage: it restores directional discrimination beyond insertion-drop alone, damps that geometry when the candidate direction is poorly resolved against its own gradient noise, and filters the Phase-1 survivors before any reduced-window refit quantity is made authoritative.

For continuity with the older local-refit notation, retain the auxiliary windowed drop

$$
\mathcal{C}_W(r;\theta) := \{\widetilde{\theta} \in \Theta_{\mathcal{O} \oplus_p m} : \widetilde{\theta}_j = (\theta \oplus_p 0)_j \ \forall j \notin W(r;t)\}
$$

and

$$
\delta E_2(r \mid \mathcal{O}, \theta) := E(\mathcal{O}, \theta) - \inf_{\widetilde{\theta} \in \mathcal{C}_W(r;\theta)} E(\mathcal{O} \oplus_p m, \widetilde{\theta}).
$$

This δE_2 quantity is retained only as an auxiliary comparator for the inherited window. It may be used for notation continuity, diagnostics, or ablation studies, but it does not define a separate selector phase, it does not enter $S_2(r;t)$, and it does not modify the Phase-II shortlist rule. Any authoritative reduced-window rerank remains deferred to Phase 3.

11.3.2 Shortlist rule

The Phase-II scoring mechanism produces the retained geometric shortlist for the later reduced-window stage. The acceptance set is $\mathcal{A}_2(t) := \{r = (m,p) \in \mathcal{R}_2 : S_2(r;t) \ge \tau_2(t)\} = \{r_{(1)}, \dots, r_{(M)}\}$, with $M := |\mathcal{A}_2(t)|$. Here S_2 is the raw geometric gain times restored novelty score. Let $s_k^{(2)}(t) := S_2(r_{(k)};t)$, such that $s_1^{(2)}(t) \ge s_2^{(2)}(t) \ge \cdots \ge s_M^{(2)}(t)$, with superscript denoting the phase scoring regime. Rate the candidate's score distinction by $u_{\mathrm{front},k}^{(2)}(t) := \frac{s_{k+1}^{(2)}(t)+\varepsilon}{s_k^{(2)}(t)+\varepsilon}$, with $k = 1, \dots, M-1$ and $0 < \eta_2 < 1$. Then

$$
k_2^\star(t) := \begin{cases} 0, & M = 0, \\ \min\Big\{ k \in \{1, \dots, \min(M-1, N_2(t)-1)\} : u_{\mathrm{front},k}^{(2)}(t) \le \eta_2 \Big\}, & \text{if this set is nonempty}, \\ \min\{M, N_2(t)\}, & \text{otherwise}, \end{cases}
$$

$$
\mathcal{D}_2(t) := \{r_{(j)} : 1 \le j \le k_2^\star(t)\}, \qquad \mathcal{S}_2(t) := \mathcal{A}_2(t) \cap \mathcal{D}_2(t) = \mathcal{D}_2(t).
$$

Of course, $\mathcal{G}^{(2)} := \pi_m(\mathcal{S}_2(t))$, and $\mathcal{P}_m^{(2)} := \{p : (m,p) \in \mathcal{S}_2(t)\}$.

11.4 Phase 3 reduced-window rerank and post-shortlist admission

This section refines the surviving Phase-2 records with the more expensive reduced-window geometry, then uses that final rerank surface for shortlist formation, batched admission, pruning, and continuation.

11.4.1 Reduced-window geometry, curvature, and trust-region gain

Phase 3 starts from the Phase-II raw quantities τ_r, g_r, h_r, $g_{\mathrm{lcb}}(r)$, $F_{\mathrm{raw}}(r)$, and $\mathcal{N}_2(r;t)$, then refines them using the inherited refit window. The coarse novelty score $\mathcal{N}_2(r;t)$ is not recomputed here; instead the candidate direction is reduced against the inherited window before the expensive rerank surface is formed.

11.4.2 Refit window, reduced geometry, and Phase-3 record family

In the record-inheritance mode used here, the Phase-3 input set is $\mathcal{R}_3(t) := \mathcal{S}_2(t)$. The inherited refit window attached to position p at selector step t is denoted $W(p;t)$, with decomposition $W(p;t) = W^{(p)}_{\mathrm{new}} \cup W^{(p)}_{\mathrm{age}}$, where $W^{(p)}_{\mathrm{new}}$ keeps the newest coordinates and $W^{(p)}_{\mathrm{age}}$ preserves the maturity-scaled set of oldest retained old coordinates defined later by the branch-local admission order. For each shortlisted record $r = (m,p)$, let $W_r := W(r;t)$ be the inherited window that would actually be refit if m were admitted at position p, and let $\{\tau_j\}_{j\in W_r}$ be the current horizontal tangents in that inherited window. Phase 3 evaluates its inherited reduced-window geometry under the later-phase controller shot law $N_{\mathrm{shot},3}(t)$.

Using the Phase-II raw tangent data, define $(Q_r)_{jk} = \Re\langle \tau_j, \tau_k\rangle$, $(q_r)_j = \Re\langle \tau_j, \tau_r\rangle$, $E_0 := E(\theta;\mathcal{O})$, $(b_r)_j := \partial_\alpha \partial_{\theta_j} E(\alpha, \delta\theta_{W_r}; r)\Big|_{\alpha=0,\,\delta\theta_{W_r}=0}$ for $j \in W_r$, and $(H_{W_r})_{jk} := \partial_{\theta_j}\partial_{\theta_k} E(\alpha, \delta\theta_{W_r}; r)\Big|_{\alpha=0,\,\delta\theta_{W_r}=0}$ for $j,k \in W_r$. The local reduced-window energy model is

$$E(\alpha, \delta\theta_{W_r}; r) \approx E_0 + g_r\alpha + \tfrac{1}{2}h_r\alpha^2 + \alpha b_r^\top \delta\theta_{W_r} + \tfrac{1}{2}\delta\theta_{W_r}^\top H_{W_r}\delta\theta_{W_r}.$$

For reduction and reduced novelty, $\mathbf{R}_r = \frac{1}{2}(H_{W_r} + H_{W_r}^\top) + \lambda_H I$, $\widetilde{h}_r = h_r - b_r^\top \mathbf{R}_r^{-1} b_r$, $F_r^{\mathrm{red}} = F_{\mathrm{raw}}(r) - 2q_r^\top \mathbf{R}_r^{-1} b_r + b_r^\top \mathbf{R}_r^{-1} Q_r \mathbf{R}_r^{-1} b_r$, $F_{r,\mathrm{safe}}^{\mathrm{red}} = \max\Big(F_r^{\mathrm{red}}, \varepsilon_{\mathrm{red}}\Big)$ with $\varepsilon_{\mathrm{red}} > 0$, $q_r^{\mathrm{red}} = q_r - Q_r\mathbf{R}_r^{-1} b_r$, and $z_r := \frac{(q_r^{\mathrm{red}})^\top (Q_r + \varepsilon_{\mathrm{nov}} I)^{-1} q_r^{\mathrm{red}}}{F_{r,\mathrm{safe}}^{\mathrm{red}}}$, $\mathcal{N}_3(r;t) := \mathrm{clip}_{[0,1]}(1 - z_r)$. The reduced trust-region gain is

$$\Delta E_{\mathrm{TR}}(r) := \max_{|\alpha| \le \rho/\sqrt{F_{r,\mathrm{safe}}^{\mathrm{red}}}} \Big[g_{\mathrm{lcb}}(r)|\alpha| - \tfrac{1}{2}\max(\widetilde{h}_r, 0)\alpha^2 \Big].$$

11.4.3 Phase-3 rerank score and effective selector

The authoritative Phase-3 rerank score is

$$S_3(r;t) := \frac{\Delta E_{\mathrm{TR}}(r)\,\chi_r\,\mathcal{N}_3(r;t)}{1 + K_3(r;t)} + \beta_{\mathrm{motif}} S_{\mathrm{motif}}(r) - \beta_{\mathrm{dup}} D_{\mathrm{dup}}(r), \qquad r \in \mathcal{R}_3(t).$$

Here S_{motif} measures motif compatibility or transferable local pattern alignment, and D_{dup} penalizes near-duplicate continuation directions. These are controller-side feature functionals normalized to comparable scale and layered on top of the explicit reduced geometric product $\Delta E_{\mathrm{TR}}(r)\,\chi_r\,\mathcal{N}_3(r;t)$; they are not part of the phase-invariant cost denominator.

The retained Phase-3 shortlist is obtained from $\mathcal{A}_3(t) := \{r = (m,p) \in \mathcal{R}_3(t) : S_3(r;t) \ge \tau_3(t)\} = \{r_{(1)}, \ldots, r_{(M)}\}$, with $M := |\mathcal{A}_3(t)|$. Let $s_k^{(3)}(t) := S_3(r_{(k)};t)$, such that $s_1^{(3)}(t) \ge s_2^{(3)}(t) \ge$

$\cdots \geq s_M^{(3)}(t)$. Rate the candidate's score distinction by $u_{\text{front},k}^{(3)}(t) := \frac{s_{k+1}^{(3)}(t)+\varepsilon}{s_k^{(3)}(t)+\varepsilon}$, with $k = 1, \ldots, M-1$ and $0 < \eta_3 < 1$. Then

$$k_3^\star(t) := \begin{cases} 0, & M = 0, \\ \min\Big\{k \in \{1, \ldots, \min(M-1, N_3(t)-1)\} : u_{\text{front},k}^{(3)}(t) \leq \eta_3\Big\}, & \text{if this set is nonempty}, \\ \min\{M, N_3(t)\}, & \text{otherwise}, \end{cases}$$

$$\mathcal{D}_3(t) := \{r_{(j)} : 1 \leq j \leq k_3^\star(t)\}, \qquad \mathcal{S}_3(t) := \mathcal{A}_3(t) \cap \mathcal{D}_3(t) = \mathcal{D}_3(t).$$

Of course, $\mathcal{G}^{(3)} := \pi_m(\mathcal{S}_3(t))$, and $\mathcal{P}_m^{(3)} := \{p : (m,p) \in \mathcal{S}_3(t)\}$.

11.4.4 Reduced-plane geometric batch selection after shortlisting

After the Phase-3 shortlisting has produced the record family $\mathcal{S}_3(t) \subseteq \mathcal{R}_3(t)$, batching is resolved by reduced-state-space geometry on a common inherited tangent plane, rather than by heuristic pairwise compatibility penalties. Support-overlap, commutation, and related proxy features may still be used as optional prescreens, but they are not part of the principled batching law below. Assume throughout that $\mathcal{S}_3(t) \neq \varnothing$; otherwise no Phase-3 batch is formed.

Near-degenerate shell and structural feasibility: let $s_t(r) := S_3(r;t)$ and $r_{\max}(t) \in \arg\max_{r \in \mathcal{S}_3(t)} s_t(r)$. Fix a near-degenerate shell parameter $\eta_{\text{nd}} \in (0,1]$ and a batch cap $B_{\max}(t) \in \mathbb{N}$. Define the near-degenerate record shell $\mathcal{N}_3(t) := \{r \in \mathcal{S}_3(t) : s_t(r) \geq \eta_{\text{nd}}\, s_t(r_{\max}(t))\}$ and the corresponding near-degenerate batch family $\mathfrak{B}_{\text{nd}}(t) := \{B \subseteq \mathcal{N}_3(t) : r_{\max}(t) \in B, \quad 1 \leq |B| \leq B_{\max}(t)\}$. Define a hard batch-structural gate $\Gamma_{\text{batch}}^{\text{struct}}(B;t) \in \{0,1\}$ to mean that the records in B are jointly insertable on the current branch/state, satisfy all hard cross-record constraints, and determine a well-defined batched scaffold update $\mathcal{O}_t \oplus B$ in the fixed canonical insertion order. The structurally feasible batch family is $\mathfrak{B}_{\text{struct}}(t) := \{B \subseteq \mathcal{S}_3(t) : \Gamma_{\text{batch}}^{\text{struct}}(B;t) = 1\}$. A controller-side transient beam fallback may also be defined on the same rerank shell.[10]

Common-window reduced geometry: write the current state as $|\psi_t\rangle := |\psi(\mathcal{O}_t, \theta_t)\rangle$, $\quad Q_{\psi_t} := I - |\psi_t\rangle\langle\psi_t|$, where Q_{ψ_t} is the horizontal projector. For any batch $B \in \mathfrak{B}_{\text{struct}}(t)$, define the common inherited refit window $W_B(t) := \bigcup_{r \in B} W_r(t)$ and the corresponding inherited horizontal tangent matrix $T_B(t) := [\, t_j(t)\,]_{j \in W_B(t)}$, $\quad t_j(t) := Q_{\psi_t}\, \partial_{\theta_j} |\psi_t\rangle$. Let $E_t(\alpha, \delta\theta_{W_B}; B)$ denote the local energy obtained by inserting the records in B with batch amplitudes $\alpha = (\alpha_r)_{r \in B}$ and varying the inherited coordinates on $W_B(t)$ by $\delta\theta_{W_B}$. For each $r \in B$, let $|\psi_{t,r}(\alpha_r)\rangle$ denote the state obtained by inserting only record r at amplitude α_r into the current scaffold while keeping inherited coordinates fixed. Its zero-initialized horizontal candidate tangent is $d_r(t) := Q_{\psi_t}\, \partial_{\alpha_r} |\psi_{t,r}(\alpha_r)\rangle\Big|_{\alpha_r = 0}$. Let $\mathsf{H}_{W_B}(t)$ denote the inherited-window Hessian block of $E_t(\alpha, \delta\theta_{W_B}; B)$ with respect to the coordinates on $W_B(t)$ at $(\alpha, \delta\theta_{W_B}) = (0,0)$, and set $M_B(t) := \frac{1}{2}\Big(\mathsf{H}_{W_B}(t) + \mathsf{H}_{W_B}(t)^\top\Big) + \lambda_H I$, $\quad \lambda_H > 0$. For each $r \in B$, define the mixed candidate-window curvature vector by $\big(b_r^{(B)}(t)\big)_j := \partial_{\alpha_r} \partial_{\theta_j} E_t(\alpha, \delta\theta_{W_B}; B)\Big|_{(\alpha, \delta\theta_{W_B}) = (0,0)}$, $\quad j \in W_B(t)$. The reduced candidate tangent after linear relaxation of the common inherited window is then $u_r^{(B)}(t) := d_r(t) - T_B(t)\, M_B(t)^{-1} b_r^{(B)}(t)$, hence the effective batch plane is $\Pi_B(t) := \operatorname{span}\{u_r^{(B)}(t) : r \in$

[10] One convenient rule is to set $\mathcal{T}_{\text{tie}}(t) := \{\, r \in \mathcal{S}_3(t) : s_t(r) \geq \max(\eta_{\text{tie}}\, s_t(r_{\max}(t)),\, s_t(r_{\max}(t)) - \delta_{\text{tie}})\,\}$. If $|\mathcal{T}_{\text{tie}}(t)| > 1$ and the controller elects not to resolve the shell by an immediate one-scaffold batch, it may retain a highest-scoring subset $\mathcal{A}_{\text{tie}}(t) \subseteq \mathcal{T}_{\text{tie}}(t)$ with $1 \leq |\mathcal{A}_{\text{tie}}(t)| \leq B_{\text{tie}}(t)$ and pass one retained alternative per record into the scaffold-beam machinery of §11.5.2.

$B\} \subseteq T_{\psi_t}\mathbb{P}(\mathcal{H})$. Its reduced Gram matrix is $\big(G_B(t)\big)_{rs} := \Re\langle u_r^{(B)}(t), u_s^{(B)}(t)\rangle, \qquad r,s \in B$, and in particular $\big(G_B(t)\big)_{rr} = F_r^{\mathrm{red},(B)}(t)$ is the reduced norm of record r in the common batch reduction. Next define the symmetrized raw candidate-candidate curvature on the same common reduction by $h_{rs}^{(B)}(t) := \frac{1}{2}\left(\partial_{\alpha_r}\partial_{\alpha_s} + \partial_{\alpha_s}\partial_{\alpha_r}\right)E_t(\alpha, \delta\theta_{W_B}; B)\big|_{(\alpha,\delta\theta_{W_B})=(0,0)}, \qquad r,s \in B$, and then the reduced curvature matrix $\big(\widetilde{H}_B(t)\big)_{rs} := h_{rs}^{(B)}(t) - \big(b_r^{(B)}(t)\big)^\top M_B(t)^{-1} b_s^{(B)}(t)$. For $r = s$, the entry $\big(\widetilde{H}_B(t)\big)_{rr}$ is the usual single-record reduced curvature in the common batch window; for $r \neq s$, the entry $\big(\widetilde{H}_B(t)\big)_{rs}$ is the mixed second-order coupling after inherited-window relaxation. The diagnostic coherences are $\mu_{\tan}(B;t) := \max_{r\neq s\in B} \frac{|(G_B(t))_{rs}|}{\sqrt{(G_B(t))_{rr}(G_B(t))_{ss}}+\varepsilon}$ and $\mu_{\mathrm{curv}}(B;t) := \max_{r\neq s\in B} \frac{|(\widetilde{H}_B(t))_{rs}|}{\sqrt{\big((\widetilde{H}_B(t))_{rr}\big)^{(+)}\big((\widetilde{H}_B(t))_{ss}\big)^{(+)}}+\varepsilon}, \qquad x^{(+)} := \max\{x, 0\}$; these may be used as optional early pruning diagnostics, but the actual batch objective is the joint reduced trust-region gain below.

Joint reduced gain and additivity: let $g_{B,\mathrm{lcb}}(t) := \big(g_{\mathrm{lcb}}(r;t)\big)_{r\in B}$, and let $\big(\widetilde{H}_B(t)\big)^{(+)}$ denote the positive-semidefinite part of $\widetilde{H}_B(t)$. The batch trust-region objective below is therefore a conservative convexified local gain model on the reduced batch plane:

$$\Delta E_B^{\mathrm{TR}}(t) := \max_{\alpha\in\mathbb{R}^{|B|},\ \alpha^\top G_B(t)\alpha \le \rho_B(t)^2} \left[g_{B,\mathrm{lcb}}(t)^\top \alpha - \frac{1}{2}\alpha^\top \big(\widetilde{H}_B(t)\big)^{(+)}\alpha \right].$$

Here $\rho_B(t) > 0$ is the Phase-3 batch trust-region radius. If record identity already includes the favorable orientation, one may additionally impose $\alpha \geq 0$ componentwise. For each $r \in B$, define the contextual single-record gain in the same common reduction by $\Delta E_{r|B}^{\mathrm{TR}}(t) := \max_{\beta\in\mathbb{R},\ \beta^2(G_B(t))_{rr}\le\rho_B(t)^2} \left[g_{\mathrm{lcb}}(r;t)\,\beta - \frac{1}{2}\big((\widetilde{H}_B(t))_{rr}\big)^{(+)}\beta^2 \right]$. Then the additivity defect is $\delta_{\mathrm{add}}(B;t) := \left[1 - \frac{\Delta E_B^{\mathrm{TR}}(t)}{\sum_{r\in B}\Delta E_{r|B}^{\mathrm{TR}}(t)+\varepsilon}\right]_+$. Small $\delta_{\mathrm{add}}(B;t)$ means that the joint reduced model is approximately additive on the common batch plane.

Batch choice and greedy implementation: the geometric admissible batch family is $\mathfrak{B}_{\mathrm{geom}}(t) := \left\{B \in \mathfrak{B}_{\mathrm{nd}}(t) \cap \mathfrak{B}_{\mathrm{struct}}(t) : \lambda_{\min}\big(G_B(t)\big) \geq \tau_{\mathrm{rank}} \frac{\operatorname{tr} G_B(t)}{|B|}, \quad \delta_{\mathrm{add}}(B;t) \le \tau_{\mathrm{add}}\right\}$. When a more aggressive pruning rule is desired, one may additionally require $\mu_{\tan}(B;t) \le \tau_{\tan}, \qquad \mu_{\mathrm{curv}}(B;t) \le \tau_{\mathrm{curv}}$. The principled post-shortlist batch choice is

$$B_t^\star \in \arg\max_{B\in\mathfrak{B}_{\mathrm{geom}}(t)} \Delta E_B^{\mathrm{TR}}(t).$$

If $\mathfrak{B}_{\mathrm{geom}}(t) = \varnothing$, no Phase-3 batch is admitted from the current shortlist. A greedy implementation avoids exhaustive search: initialize $B_t^{(1)} := \{r_{\max}(t)\}$. Given a current batch $B_t^{(k)}$, define the admissible add-on family $\mathcal{A}_k(t) := \left\{r \in \mathcal{N}_3(t) \setminus B_t^{(k)} : B_t^{(k)} \cup \{r\} \in \mathfrak{B}_{\mathrm{nd}}(t) \cap \mathfrak{B}_{\mathrm{struct}}(t)\right\}$. For each $r \in \mathcal{A}_k(t)$, define the marginal joint gain $\Delta_{\mathrm{marg}}(r \mid B_t^{(k)};t) := \Delta E_{B_t^{(k)}\cup\{r\}}^{\mathrm{TR}}(t) - \Delta E_{B_t^{(k)}}^{\mathrm{TR}}(t)$. Choose $r_k^{\mathrm{best}}(t) \in \arg\max_{r\in\mathcal{A}_k(t)} \Delta_{\mathrm{marg}}(r \mid B_t^{(k)};t)$ subject to $\lambda_{\min}\big(G_{B_t^{(k)}\cup\{r\}}(t)\big) \geq \tau_{\mathrm{rank}} \frac{\operatorname{tr} G_{B_t^{(k)}\cup\{r\}}(t)}{|B_t^{(k)}|+1}, \qquad \delta_{\mathrm{add}}(B_t^{(k)} \cup \{r\};t) \le \tau_{\mathrm{add}}$, and, when enabled, $\mu_{\tan}(B_t^{(k)} \cup \{r\};t) \le \tau_{\tan}, \qquad \mu_{\mathrm{curv}}(B_t^{(k)} \cup \{r\};t) \le \tau_{\mathrm{curv}}$. If no such r exists, or if the best feasible marginal gain is nonpositive, stop and set $B_t^\star := B_t^{(k)}$. Otherwise append the best add-on, $B_t^{(k+1)} := B_t^{(k)} \cup \{r_k^{\mathrm{best}}(t)\}$, and continue until no further feasible positive-gain append exists or until $|B_t^{(k)}| = B_{\max}(t)$. The deployed Phase-3 action is the final batch $B_t^\star$. If $|B_t^\star| = 1$, this reduces to the usual single-record admission. If $|B_t^\star| > 1$, the controller inserts the batch on the current branch in the canonical batch order and performs the common-window refit on $W_{B_t^\star}(t)$.

11.5 Post m_p Application Entails Windowed-Refitting

In ADAPT-VQE the variational problem is $\min_{\boldsymbol{\theta}} E_L(\boldsymbol{\theta})$ with $E_L(\boldsymbol{\theta}) = \langle\phi_0|(\prod_{\ell=L}^{1} e^{i\theta_\ell G_\ell})H(\prod_{\ell=1}^{L} e^{-i\theta_\ell G_\ell})|\phi_0\rangle$, and the measurement bottleneck arises because one iteration must estimate not only E_L but also pool-selection scores $s_\mu(\boldsymbol{\theta}) = |\langle\phi_0|(\prod_{\ell=L}^{1} e^{i\theta_\ell G_\ell}), i[H, A_\mu], (\prod_{\ell=1}^{L} e^{-i\theta_\ell G_\ell})|\phi_0\rangle|$ for $\mu \in \mathcal{P}$, reoptimization gradients $\partial_{\theta_j} E_L(\boldsymbol{\theta})$, and, for geometry-aware updates, metric entries $M_{jk}(\boldsymbol{\theta}) = \Re\langle\partial_{\theta_j}[(\prod_{\ell=1}^{L} e^{-i\theta_\ell G_\ell})|\phi_0\rangle]|\partial_{\theta_k}[(\prod_{\ell=1}^{L} e^{-i\theta_\ell G_\ell})|\phi_0\rangle]\rangle$, so at precisions $\epsilon_s, \epsilon_g, \epsilon_M$ the shot cost obeys $N_{\text{iter}} \gtrsim \sum_{\mu\in\mathcal{P}} \mathrm{Var}(\hat{s}_\mu)/\epsilon_s^2 + \sum_{j=1}^{L} \mathrm{Var}(\widehat{\partial_{\theta_j} E_L})/\epsilon_g^2 + \sum_{j,k=1}^{L} \mathrm{Var}(\hat{M}_{jk})/\epsilon_M^2$; the proposed strategy is to choose disjoint sets $F, A \subset 1, \ldots, L$ with $F \cup A = 1, \ldots, L$, impose $\dot{\theta}_j = 0$ for $j \in F$, and optimize only $j \in A$, which reduces the reparameterization sector to $\sum_{j\in A} \mathrm{Var}(\widehat{\partial_{\theta_j} E_L})/\epsilon_g^2 + \sum_{j,k\in A} \mathrm{Var}(\hat{M}_{jk})/\epsilon_M^2$ and therefore removes the growth coming from repeatedly updating old coordinates, but it leaves the pool-screening term $\sum_{\mu\in\mathcal{P}} \mathrm{Var}(\hat{s}_\mu)/\epsilon_s^2$ unchanged and it does not shorten the physical circuit, since every shot still executes the full depth-L scaffold with entangling-gate count $N_{2q}(L)$ and runtime $t_{\text{circ}}(L)$, so hardware fidelity still degrades roughly as $e^{-t_{\text{circ}}(L)/T_1} e^{-t_{\text{circ}}(L)/T_2} (1 - \varepsilon_{2q})^{N_{2q}(L)}$; consequently the freezing strategy cures only the reoptimization component of the ADAPT bottleneck, and the limiting cost after that is either the unchanged operator-pool screening overhead or, if screening is separately controlled, the conventional VQE bottleneck of resolving E_L on a deep noisy ansatz, with shots scaling as $N_E \gtrsim \mathrm{Var}(H_{\text{meas}} \mid \prod_{\ell=1}^{L} e^{-i\theta_\ell G_\ell}, \phi_0, \text{noise})/(\Delta E)^2$ to distinguish an energy change ΔE.

Let the length-(m) scaffold be $U^{(m)}(\boldsymbol{\theta}) = \prod_{j=1}^{m} e^{-i\theta_j G_j}$, where (G_j) is the (j)-th appended generator and (θ_j) its coefficient. The reparameterization motif is to optimize only an active subset $(\mathcal{A}_m \subseteq 1, \ldots, m)$, chosen by an activity functional $a_j^{(m)} := \lambda_{\text{rec}}\, r_j^{(m)} + \lambda_{\text{mob}}\, v_j^{(m)} - \lambda_{\text{age}}\, \phi(m-j) - \lambda_{\text{stag}}\, s_j^{(m)}$, and then set $(\mathcal{A}_m = \{j : a_j^{(m)} > \tau_m\})$. Here $(r_j^{(m)})$ is a recency weight, large for recently appended generators; $(v_j^{(m)} := \sum_{k=j}^{m-1} |\Delta\theta_j^{(k)}|)$ is a lifetime mobility measure, large for parameters that have historically moved substantially; $(\phi(m-j))$ is an age penalty, biasing against early-added generators; and $(s_j^{(m)})$ is a stagnation indicator, large when a parameter has remained nearly inert over recent refits. Thus the inclusion bias is toward recent and historically labile generators, while the exclusion bias is toward old and typically stagnant generators.

This is the same scaffold language used above: $U^{(m)}(\boldsymbol{\theta})$ is the scaffold $\mathcal{O}$, the active set $\mathcal{A}_m$ plays the role of the inherited refit window $W(r;t)$, the frozen set is $\{1, \ldots, n+1\} \setminus W(r;t)$, and the activity functional $a_j^{(m)}$ is the same maturity-facing bookkeeping later summarized by the eligible set $\mathcal{M}_t$.

Concomitantly, as scaffold maturity increases, the controller shot laws $N_{\text{shot},1}(t)$ and $N_{\text{shot},k}(t)$ should also increase so as to preserve the decision-level signal-to-noise ratio on the active block. Formally, if δ_t denotes a typical useful signal scale among the unfrozen directions and σ_t the corresponding estimator spread, then one chooses the phase-appropriate shot law so that $\mathrm{SNR}_t \sim \frac{\delta_t \sqrt{N_{\text{shot},1}(t)}}{\sigma_t} \geq \kappa$ in Phase 1 and $\mathrm{SNR}_{t,k} \sim \frac{\delta_{t,k}\sqrt{N_{\text{shot},k}(t)}}{\sigma_{t,k}} \geq \kappa$ for $k \in \{2, 3\}$ in later phases. Hence the control law is: as scaffold maturity grows, shrink the reoptimized subspace by freezing archaic or stagnant directions while increasing shots on the surviving active directions to maintain resolvable updates.

11.5.1 Branch-local mature-scaffold pruning: permissibility, frozen-ablation ranking, and exact rollback

All quantities in this subsection are branch-local. On a single active branch we suppress the branch label. Let $(\mathcal{O}_t^-, \theta_t^-, E_t^-)$ be the committed branch state immediately before the accepted Phase-3

payload $\mathcal{A}_t^\star$ is applied, and let $(\mathcal{O}_t^+, \theta_t^+, E_t^+)$ be the committed post-admission state after the ordinary local refit on that same branch has completed. The admitted-step gain is $\Delta_{\mathrm{adm}}(t) := E_t^- - E_t^+$.

Each committed coordinate of $\mathcal{O}_t^+$ carries the metadata $(m_j, \iota_j, r_j, \kappa_j, c_j)$: generator label m_j, branch-local admission step ι_j, admission record r_j, stored selector burden $\kappa_j := K_3(r_j; \iota_j)$, and prune-cooldown counter $c_j(t) \in \{0, 1, \ldots, c_{\mathrm{cool}}\}$, transported with that coordinate under later scaffold edits. Here $t_{\mathrm{first}}(m_j)$ denotes the first time the generator m_j was ever seen in the run.

Use the maturity-increasing prune-pressure law $\lambda_{\mathrm{prune}}(t) := \lambda_{\mathrm{prune}}^{\min} + (\lambda_{\mathrm{prune}}^{\max} - \lambda_{\mathrm{prune}}^{\min}) g_t$, with $0 \le \lambda_{\mathrm{prune}}^{\min} \le \lambda_{\mathrm{prune}}^{\max}$. Let the baseline gain-resolution scale be $\Delta_{\mathrm{scr}}(t) := \max\big(\tau_{\mathrm{use},t}, \mathrm{EWMA}_{\mathrm{recent}}(\Delta_{\mathrm{adm}}), \Delta_{\mathrm{num}}\big)$, use the noise-aware screening scale $\Delta_{\mathrm{scr}}^{\mathrm{noise}}(t) := (1 + u_{\sigma,3}(t))\Delta_{\mathrm{scr}}(t)$, and define the admitted-gain signal-to-noise ratio by $\mathrm{SNR}_{\mathrm{adm}}(t) := \Delta_{\mathrm{adm}}(t)/(\Delta_{\mathrm{scr}}^{\mathrm{noise}}(t) + \varepsilon)$. A prune pass may open only after an accepted admission and its ordinary local refit have already committed. Let $\mathcal{T}_{\mathrm{chk}} \subseteq \mathbb{N}$ be the scheduled checkpoint set, define $C_t^{\mathrm{perm}} := (u_{\mathrm{sat}}(t) \ge \tau_{\mathrm{med}}) \wedge \Big(\mathrm{SNR}_{\mathrm{adm}}(t) \le \kappa_{\mathrm{pr}} \ \vee \ t \in \mathcal{T}_{\mathrm{chk}}\Big)$, and $\Gamma_{\mathrm{prune}}^{\mathrm{perm}}(t) := \mathbf{1}[\text{accepted admission at } t]\mathbf{1}[\text{post-admission local refit completed}]\mathbf{1}[|\mathcal{O}_t^+| \ge 2]\mathbf{1}[C_t^{\mathrm{perm}}]$. Thus pruning is a maturity-controlled maintenance action that opens only when the scaffold is already in the mature regime and the admitted gain is no longer cleanly resolved above the noise-aware screening floor, or when a maturity-gated checkpoint explicitly requests a simplification test.

Newly admitted coordinates are protected for ℓ_{protect} admitted steps via $\mathcal{P}_{\mathrm{protect}}(t) := \{\, j : t - \iota_j < \ell_{\mathrm{protect}} \,\}$. Let $s_j^{\mathrm{stag}}(t)$ denote the branch-local t-indexed lift of the earlier coordinate stagnation indicator $s_j^{(m)}$, with larger values meaning less recent parameter motion and therefore greater local stagnation. The mature prune-eligible set is

$$\mathcal{M}_t := \Big\{ j : \ t - t_{\mathrm{first}}(m_j) \ge A_{\mathrm{stale}}, \ s_j^{\mathrm{stag}}(t) \ge s_{\mathrm{stale}}, \ j \notin \mathcal{P}_{\mathrm{protect}}(t), \ c_j(t) = 0 \Big\}.$$

Hence prune eligibility is branch-local and conservative: old by first-seen age, stale by recent inactivity, not recently admitted, and not under temporary cooldown.

Small angle is used only as a cheap nomination device. If $\mathcal{M}_t \neq \varnothing$, set $\theta_{\mathrm{small}}(t) := \max\big(\theta_{\mathrm{abs}}, c_\theta \operatorname{median}_{j \in \mathcal{M}_t} |\operatorname{wrap}(\theta_j(t))|\big)$ and $\mathcal{J}_\angle(t) := \{\, j \in \mathcal{M}_t : |\operatorname{wrap}(\theta_j(t))| \le \theta_{\mathrm{small}}(t) \,\}$; let $\mathcal{P}_{\mathrm{probe}}(t) \subseteq \mathcal{J}_\angle(t)$ be the set of the $\min\{K_p, |\mathcal{J}_\angle(t)|\}$ smallest values of $|\operatorname{wrap}(\theta_j(t))|$ over $j \in \mathcal{J}_\angle(t)$. If $\mathcal{M}_t = \varnothing$, set $\mathcal{P}_{\mathrm{probe}}(t) = \varnothing$. Thus old/stale and small-angle coordinates are nominated cheaply, but neither condition is itself an admissible proof of redundancy.

For each $j \in \mathcal{P}_{\mathrm{probe}}(t)$, let $\mathcal{O}_{t,-j}^+$ denote the scaffold obtained from $\mathcal{O}_t^+$ by deleting coordinate j, and let $(\theta_t^+)_{-j}$ be θ_t^+ with the same entry removed. The frozen-ablation energy and immediate regression are $E_{-j}^{\mathrm{frz}} := E(\mathcal{O}_{t,-j}^+, (\theta_t^+)_{-j})$ and $L_j^{\mathrm{frz}}(t) := E_{-j}^{\mathrm{frz}} - E_t^+$. Use the cheap prune score

$$P_j^{\mathrm{pr}}(t) := \frac{[L_j^{\mathrm{frz}}(t)]_+}{1 + \kappa_j}, \qquad j \in \mathcal{P}_{\mathrm{probe}}(t),$$

and, when $\mathcal{P}_{\mathrm{probe}}(t) \neq \varnothing$, choose

$$j_t^\star := \arg \min_{j \in \mathcal{P}_{\mathrm{probe}}(t)} P_j^{\mathrm{pr}}(t).$$

Thus the cheap ranking reuses the same K-style burden logic already used for admission: low retained frozen-ablation utility relative to stored selector burden means good prune-test priority, so smaller $P_j^{\mathrm{pr}}(t)$ is better.

Before the authoritative trial, store the exact rollback snapshot $\Xi_t^{\mathrm{pr}} := (\mathcal{O}_t^+, \theta_t^+, E_t^+, \{c_i(t)\}_i, \text{current local optimiz}$ Only one generator is tested per prune pass. If $\Gamma_{\mathrm{prune}}^{\mathrm{perm}}(t) = 0$ or $\mathcal{P}_{\mathrm{probe}}(t) = \varnothing$, no prune trial is launched and the committed branch state remains $(\mathcal{O}_t^+, \theta_t^+, E_t^+)$.

For the authoritative trial on $j_t^\star$, let $n_t^+ := |\mathcal{O}_t^+|$, $\omega_{\text{eff}}^{(-j)} := \max\{1, \min(\omega_{\text{pr}}, n_t^+ - 1)\}$, $\ell_j := \max\left\{1, \min\left(j - \left\lfloor \frac{\omega_{\text{eff}}^{(-j)}-1}{2} \right\rfloor,\ n_t^+ - \omega_{\text{eff}}^{(-j)}\right)\right\}$, $W_{\text{loc}}(j) := \{\ell_j, \ldots, \ell_j + \omega_{\text{eff}}^{(-j)} - 1\}$, $C_{\text{old}}^{(-j)} := \{1, \ldots, n_t^+ - 1\} \setminus W_{\text{loc}}(j)$, and $n_{\text{old}}^{\text{pr}}(t, j) := \left\lceil \rho_W(t)\, |C_{\text{old}}^{(-j)}| \right\rceil$. Let $W_{\text{age}}^{(-j)}(t)$ be the set of the $n_{\text{old}}^{\text{pr}}(t, j)$ oldest surviving coordinates in $C_{\text{old}}^{(-j)}$ under the stored admission-order values ι_i. The prune-refit window is

$$W_{\text{pr}}(j; t) := W_{\text{loc}}(j) \cup W_{\text{age}}^{(-j)}(t).$$

Hence the prune refit window contracts automatically as maturity increases, because $\rho_W(t)$ decreases as useful runway is exhausted.

The authoritative post-ablation refit is

$$\theta_t^{(-j)} := \arg\min_{\widetilde{\theta} \in \Theta_{\mathcal{O}_{t,-j}^+}} \left\{ E(\mathcal{O}_{t,-j}^+, \widetilde{\theta}) : \widetilde{\theta}_i = ((\theta_t^+)_{-j})_i \ \forall i \notin W_{\text{pr}}(j; t) \right\},$$

with corresponding post-refit energy $E_t^{(-j)} := E(\mathcal{O}_{t,-j}^+, \theta_t^{(-j)})$. This ablation-plus-local-refit result is the authoritative prune test. No extra geometric score is required at this stage.

Define the retained admitted-step gain after removing j by $\Delta_{\text{keep}}^{(-j)}(t) := E_t^- - E_t^{(-j)}$. Accept the prune of $j_t^\star$ iff $E_t^{(-j_t^\star)} \le E_t^+ + \Delta_{\text{safe}}$ and $\Delta_{\text{keep}}^{(-j_t^\star)}(t) \ge \eta_{\text{ret}} \Delta_{\text{adm}}(t)$. Thus the final accept rule is recoverability-based: the reduced scaffold may be kept only when the ablation-plus-refit energy regression is numerically small and the pruned scaffold still retains a controlled fraction of the gain produced by the last admitted payload. The burden term κ_j is used only for cheap ranking; the final accept rule remains energy-based and conservative.

If the prune is accepted, commit $(\mathcal{O}_{t,-j_t^\star}^+, \theta_t^{(-j_t^\star)}, E_t^{(-j_t^\star)})$ as the new branch state and delete the metadata carried by coordinate $j_t^\star$. If the prune is rejected, restore the branch state exactly to Ξ_t^{pr}, set $c_{j_t^\star}(t) \leftarrow c_{\text{cool}}$, and launch no second prune trial at the same selector step. On later admitted steps the cooldown counter decrements by one until it reaches zero. Hence prune failure never invalidates the already committed admission, and rollback is exact.

11.5.2 Beam-adapt over scaffold alternatives: branch state, pruning, and effective selector

In this section beam-adapt is **purely structural**: each branch carries an alternative ADAPT scaffold together with its locally refit amplitudes. There is no time checkpoint, probe rollout, or projected-dynamics integral in this section; those belong to the separate time-dynamics beam surface developed later.

The cleanest way to read a live scaffold branch is

$$\begin{aligned}\mathfrak{b} = \big(&\text{identity/history; current scaffold and amplitudes;}\\ &\text{current energy/depth; cumulative selector totals/controller telemetry/status}\big).\end{aligned}$$

More explicitly, write

$$\mathfrak{b} = (\text{id}_b, \pi_b, \mathcal{H}_b, \mathcal{O}_b, \theta_b, E_b, d_b, S_b^{\text{cum}}, K_b^{\text{cum}}, \mathcal{T}_b^{\text{ctrl}}, \text{status}_b, \text{term}_b),$$

with $\mathcal{H}_b = (r_{b,1}, \ldots, r_{b,d_b})$ and $r_{b,j} = (m_{b,j}^{\text{adm}}, p_{b,j}^{\text{adm}})$. Here id_b and π_b are the branch and parent identifiers, $\mathcal{H}_b$ is the ordered history of admitted records, $\mathcal{O}_b$ is the scaffold currently carried by branch b, θ_b is the refit parameter vector on that scaffold, $E_b = E(\theta_b; \mathcal{O}_b)$ is the current variational

energy, d_b is the branch ADAPT depth, $S_b^{\mathrm{cum}} = \sum_{j=1}^{d_b} s_{b,j}$ is the cumulative admitted selector value, $K_b^{\mathrm{cum}} = \sum_{j=1}^{d_b} K_{b,j}^{\mathrm{sel}}$ is the cumulative admitted selector burden, $\mathcal{T}_b^{\mathrm{ctrl}}$ stores branch-local controller telemetry, and $(\mathrm{status}_b, \mathrm{term}_b)$ record live/terminal status and termination label. The increment $K_{b,j}^{\mathrm{sel}}$ is the selector-cost contribution attached to the j-th admission on branch b; for a newly materialized child this contribution is the branch-local Phase-3 cost $K_3(r;\mathfrak{b})$. Thus K_b^{cum} is cumulative selector-side cost bookkeeping rather than a raw hardware count.

If beam search is entered from an incumbent scaffold $(\mathcal{O}^{(0)}, \theta^{(0)})$, the root branch is $\mathfrak{b}_{\mathrm{root}} = (\mathrm{id}_0, \varnothing, \mathcal{H}^{(0)}, \mathcal{O}^{(0)}, \theta^{(0)}, E(\theta^{(0)}; \mathcal{O}^{(0)}), d_0, S_{\mathrm{root}}^{\mathrm{cum}}, K_{\mathrm{root}}^{\mathrm{cum}}, \mathcal{T}_{\mathrm{root}}^{\mathrm{ctrl}}, \mathrm{frontier}, \varnothing)$, with $d_0 = |\mathcal{H}^{(0)}|$, $S_{\mathrm{root}}^{\mathrm{cum}} = \sum_{j=1}^{d_0} s_j^{(0)}$, and $K_{\mathrm{root}}^{\mathrm{cum}} = \sum_{j=1}^{d_0} K_j^{\mathrm{sel},(0)}$, where $K_j^{\mathrm{sel},(0)}$ denotes the selector-cost contribution attached to the j-th historical admission on the incumbent scaffold. If the beam is started from a fresh scaffold, then $\mathcal{H}^{(0)} = \varnothing$, $d_0 = 0$, and both cumulative sums vanish.

For clock and budget, set $t_b := d_b$ and $D_{\mathrm{left}}(\mathfrak{b}) := d_{\max} - d_b$. Now apply the branch lift rule: for any live-state quantity X from Phase 3 whose definition is written in terms of $(\mathcal{H}, \mathcal{O}, \theta, t, \mathcal{T}^{\mathrm{ctrl}})$, define $X(\mathfrak{b}) := X\big|_{(\mathcal{H},\mathcal{O},\theta,t,\mathcal{T}^{\mathrm{ctrl}}) \mapsto (\mathcal{H}_b,\mathcal{O}_b,\theta_b,t_b,\mathcal{T}_b^{\mathrm{ctrl}})}$; for any record-local quantity $X(r;t)$, define $X(r;\mathfrak{b}) := X(r;t)\big|_{(\mathcal{H},\mathcal{O},\theta,t,\mathcal{T}^{\mathrm{ctrl}}) \mapsto (\mathcal{H}_b,\mathcal{O}_b,\theta_b,t_b,\mathcal{T}_b^{\mathrm{ctrl}})}$. In particular, $R(\mathfrak{b})$, $u_{\mathrm{sat}}(\mathfrak{b})$, $\widehat{N}_{\mathrm{rem}}(\mathfrak{b})$, $K_3(r;\mathfrak{b})$, and $g_{\mathrm{lcb}}(r;\mathfrak{b})$ are branch-lifted evaluations of their previously defined live-state counterparts.

The Phase-3 chain is branch-local by construction: all of its dependencies factor through the state tuple $(\mathcal{O}_b, \theta_b, t_b, \mathcal{T}_b^{\mathrm{ctrl}})$. Hence each live branch induces its own raw candidate-position surface $\mathcal{R}_b^{\mathrm{raw}}$, its own branch-local shortlist $\mathcal{S}_3(\mathfrak{b})$, and its own retained admission set $\mathcal{A}_b$.

For stage and enabled family, let $\eta_b \in \{\mathrm{core/seed}, \mathrm{residual}\}$ and

$$\mathcal{G}_{\eta_b}^{(3)}(\mathfrak{b}) = \begin{cases} \mathcal{G}_{\mathrm{core/seed}}^{(3)}(\mathfrak{b}), & \eta_b = \mathrm{core/seed}, \\ \mathcal{G}_{\mathrm{residual}}^{(3)}(\mathfrak{b}), & \eta_b = \mathrm{residual}. \end{cases}$$

Let $\mathcal{P}_{\mathrm{residual}}(\mathfrak{b}) \subseteq \mathcal{G}^{(3)}$ denote the residual-only generator family on branch b. Then the controller-stage gate and branch-local available Phase-3 generator family are

$$\Gamma_{\mathrm{stage},b}(m) = \begin{cases} 1, & \eta_b = \mathrm{residual}, \\ 1, & \eta_b = \mathrm{core/seed} \text{ and } m \notin \mathcal{P}_{\mathrm{residual}}(\mathfrak{b}), \\ 0, & \eta_b = \mathrm{core/seed} \text{ and } m \in \mathcal{P}_{\mathrm{residual}}(\mathfrak{b}), \end{cases} \qquad \mathcal{G}_{\mathrm{avail}}^{(3)}(\mathfrak{b}) = \{m \in \mathcal{G}^{(3)} : \Gamma_{\mathrm{stage},b}(m) = 1\}.$$

For scaffold length, append slot, and active probe window, let $n_b = |\mathcal{O}_b|$, $p_{\mathrm{app}}(\mathfrak{b}) = n_b$, and $W_{\mathrm{act}}(\mathfrak{b}) \subseteq \{0, \ldots, n_b - 1\}$. The ordered probe list is $\mathcal{L}_{\mathrm{probe}}(\mathfrak{b}) := \big(p_{\mathrm{app}}(\mathfrak{b}), 0, W_{\mathrm{act}}(\mathfrak{b})\big)$, and the distinct retained probe positions are $\mathcal{P}_{\mathrm{probe}}(\mathfrak{b}) = \mathrm{Head}_{M_{\mathrm{probe}}}\Big(\mathrm{Dedup}_{\mathrm{ord}}\big(\mathcal{L}_{\mathrm{probe}}(\mathfrak{b})\big)\Big) \subseteq \{0, \ldots, n_b\}$. For symmetry and admissible positions, let

$$\Gamma_{\mathrm{sym},b}(m,p) = \begin{cases} 0, & \text{the symmetry audit for } (m,p;\mathcal{O}_b) \text{ returns a hard violation}, \\ 1, & \text{otherwise}, \end{cases}$$

so the branch-local raw candidate-position surface is

$$\mathcal{R}_b^{\mathrm{raw}} := \{\, r = (m,p) : m \in \mathcal{G}_{\mathrm{avail}}^{(3)}(\mathfrak{b}),\ p \in \mathcal{P}_{\mathrm{probe}}(\mathfrak{b}),\ \Gamma_{\mathrm{sym},b}(m,p) = 1 \,\}.$$

The resulting local shortlist and retained admission set are

$$\mathcal{S}_3(\mathfrak{b}) := \mathrm{Top}_{N_3(t_b)}(S_3(\cdot;\mathfrak{b}); \mathcal{R}_b^{\mathrm{raw}}), \qquad \mathcal{A}_b := \mathrm{Top}_{K_b}(S_3(\cdot;\mathfrak{b}); \{r \in \mathcal{S}_3(\mathfrak{b}) : S_3(r;\mathfrak{b}) > 0\}).$$

When a branch-local rerank shell is judged tie-degenerate, the retained admission set may be replaced by a transient tie-shell selector.[11]

(ii) Materialize either stop or one new admission. The proposal family is $\mathcal{Q}(b) = \{\text{stop}\} \cup \mathcal{A}_b$. The symbol stop means "terminate this scaffold as-is," while each $r = (m, p) \in \mathcal{A}_b$ means "insert one new generator into this branch and refit." If $\mathcal{O}_b = (m_{b,1}, \ldots, m_{b,n_b})$ and $0 \le p \le n_b$, then the scaffold insertion map is $\mathrm{Insert}(\mathcal{O}_b, m, p) = (m_{b,1}, \ldots, m_{b,p}, m, m_{b,p+1}, \ldots, m_{b,n_b})$, the insertion index transport is $I_p(j) = \begin{cases} j, & 1 \le j \le p, \\ j+1, & p < j \le n_b, \end{cases}$, and the zero-lifted parameter vector is $\theta_b^{\uparrow(r)} = (\theta_{b,1}, \ldots, \theta_{b,p}, 0, \theta_{b,p+1}, \ldots, \theta_{b,n_b})$.

For contiguous inserted-window cap ω_{new}, let $\omega_{\text{eff}}^{(p)} := \max\{1, \min(\omega_{\text{new}}, n_b + 1)\}$ and $\ell_p := \max\left\{1, \min\left(p + 1 - \left\lfloor \frac{\omega_{\text{eff}}^{(p)} - 1}{2} \right\rfloor,\ n_b + 2 - \omega_{\text{eff}}^{(p)}\right)\right\}$. Then $W_{\text{new}}^{(p)} = \{\ell_p,\ \ell_p + 1,\ \ldots,\ \ell_p + \omega_{\text{eff}}^{(p)} - 1\}$, so $|W_{\text{new}}^{(p)}| = \omega_{\text{eff}}^{(p)}$ and $p + 1 \in W_{\text{new}}^{(p)}$. Let the retained old-coordinate fraction shrink with maturity according to $\rho_W(t) := \rho_W^{\min} + (\rho_W^{\max} - \rho_W^{\min})e_t$, with $0 \le \rho_W^{\min} \le \rho_W^{\max} \le 1$, and define the old-coordinate complement by $C_{\text{old}}^{(p)} := \{1, \ldots, n_b + 1\} \setminus W_{\text{new}}^{(p)}$, so $|C_{\text{old}}^{(p)}| = n_b + 1 - \omega_{\text{eff}}^{(p)}$. Then the retained old-coordinate count is $n_{\text{old}}(t, p) := \left\lceil \rho_W(t)\, |C_{\text{old}}^{(p)}| \right\rceil = \left\lceil \rho_W(t)\, (n_b + 1 - \omega_{\text{eff}}^{(p)}) \right\rceil$. For each current scaffold coordinate $j \in \{1, \ldots, n_b\}$, let $a_{b,j}^{\text{adm}}$ denote the admission step at which that coordinate entered the current scaffold on branch b, with smaller values meaning older coordinates, and let the insertion-lifted admission-order vector be $a_b^{\uparrow(r)} := (a_{b,1}^{\text{adm}}, \ldots, a_{b,p}^{\text{adm}}, +\infty, a_{b,p+1}^{\text{adm}}, \ldots, a_{b,n_b}^{\text{adm}})$, so the newly inserted slot is never selected as an old coordinate. The retained old-coordinate window is then $W_{\text{age}}^{(p)} := \{\, j \in C_{\text{old}}^{(p)} : \#\{\, i \in C_{\text{old}}^{(p)} : (a_b^{\uparrow(r)})_i < (a_b^{\uparrow(r)})_j \,\} < n_{\text{old}}(t, p) \,\}$, namely the set of the $n_{\text{old}}(t, p)$ oldest old coordinates outside $W_{\text{new}}^{(p)}$, and hence the inherited refit window on the inserted scaffold is $W_r = W(r; t) := W_{\text{new}}^{(p)} \cup W_{\text{age}}^{(p)}$.

Then, for $r = (m, p)$ and $\mathcal{O}_b \oplus_p m := \mathrm{Insert}(\mathcal{O}_b, m, p)$, the branch-local refit map is $\mathrm{Refit}_{W_r}(\theta_b; r) := \arg\min_{\widetilde{\theta} \in \Theta_{\mathcal{O}_b \oplus_p m}} \left\{ E(\mathcal{O}_b \oplus_p m, \widetilde{\theta}) :\ \widetilde{\theta}_j = (\theta_b^{\uparrow(r)})_j\ \forall j \notin W_r \right\}$.

The transfer map is $\mathcal{T}(\mathfrak{b}, \text{stop}) = \mathfrak{b}'$ with $\mathcal{H}_{b'} = \mathcal{H}_b$, $\mathcal{O}_{b'} = \mathcal{O}_b$, $\theta_{b'} = \theta_b$, $E_{b'} = E_b$, $d_{b'} = d_b$, $S_{b'}^{\text{cum}} = S_b^{\text{cum}}$, $K_{b'}^{\text{cum}} = K_b^{\text{cum}}$, $\text{status}_{b'} = \text{terminal}$, and $\text{term}_{b'} = \begin{cases} \text{empty}, & \mathcal{A}_b = \varnothing, \\ \text{stop}, & \mathcal{A}_b \neq \varnothing. \end{cases}$ For $r = (m, p)$, $\mathcal{T}(\mathfrak{b}, r) = \mathfrak{b}'$ with $\mathcal{O}_{b'} = \mathrm{Insert}(\mathcal{O}_b, m, p)$, $\theta_{b'} = \mathrm{Refit}_{W_r}(\theta_b; r)$, $\mathcal{H}_{b'} = (\mathcal{H}_b, r)$, $E_{b'} = E(\theta_{b'}; \mathcal{O}_{b'})$, $d_{b'} = d_b + 1$, $S_{b'}^{\text{cum}} = S_b^{\text{cum}} + S_3(r; \mathfrak{b})$, $K_{b'}^{\text{cum}} = K_b^{\text{cum}} + K_3(r; \mathfrak{b})$, $\text{status}_{b'} = \begin{cases} \text{terminal}, & d_{b'} \ge d_{\max} \text{ or } \mathcal{A}_{b'} = \varnothing, \\ \text{frontier}, & d_{b'} < d_{\max} \text{ and } \mathcal{A}_{b'} \neq \varnothing, \end{cases}$, and $\text{term}_{b'} = \begin{cases} \text{depth_cap}, & d_{b'} \ge d_{\max}, \\ \text{empty}, & d_{b'} < d_{\max} \text{ and } \mathcal{A}_{b'} = \varnothing, \\ \varnothing, & d_{b'} < d_{\max} \text{ and } \mathcal{A}_{b'} \neq \varnothing. \end{cases}$ where $\mathcal{A}_{b'}$ is recomputed from step (i) on the child branch itself. So a non-stop child is exactly one more ADAPT admission applied to the parent scaffold, followed by the inherited-window refit already defined above, and its cumulative K-coordinate is updated by the admitted branch-local Phase-3 selector burden $K_3(r; \mathfrak{b})$. If $\mathcal{A}_b = \varnothing$, then $\mathcal{Q}(b) = \{\text{stop}\}$ and the branch contributes only a terminal child.

(iii) Classify, deduplicate, and prune the scaffold children. The frontier and terminal descendants of branch b are $\mathfrak{C}_b^{\text{front}} = \{\mathfrak{b}' : \mathfrak{b}' = \mathcal{T}(\mathfrak{b}, r),\ r \in \mathcal{A}_b,\ \text{status}_{b'} = \text{frontier}\}$ and $\mathfrak{C}_b^{\text{term}} = \{\mathfrak{b}' : \mathfrak{b}' = \mathcal{T}(\mathfrak{b}, q),\ q \in \mathcal{Q}(b),\ \text{status}_{b'} = \text{terminal}\}$. Two branches are treated as duplicates

[11] For example, if $r_{b,\max} \in \arg\max_{r \in \mathcal{S}_3(\mathfrak{b})} S_3(r; \mathfrak{b})$, one may define $\mathcal{T}_{\text{tie}}(\mathfrak{b}) := \{\, r \in \mathcal{S}_3(\mathfrak{b}) : S_3(r; \mathfrak{b}) \ge \max(\eta_{\text{tie}}\, S_3(r_{b,\max}; \mathfrak{b}),\ S_3(r_{b,\max}; \mathfrak{b}) - \delta_{\text{tie}}) \,\}$ and replace $\mathcal{A}_b$ by a retained subset $\mathcal{A}_b^{\text{tie}} \subseteq \mathcal{T}_{\text{tie}}(\mathfrak{b})$ with $1 \le |\mathcal{A}_b^{\text{tie}}| \le B_{\text{tie}}(t_b)$, chosen by descending $S_3(\cdot; \mathfrak{b})$.

when they represent the same scaffold and essentially the same refit point, for example under the fingerprint $\mathrm{rnd}_{10}(x) := 10^{-10}\,\mathrm{round}(10^{10}x)$, $\mathrm{round}_{10}(\theta_b) := \big(\mathrm{rnd}_{10}(\theta_{b,1}), \ldots, \mathrm{rnd}_{10}(\theta_{b,n_b})\big)$, and $\mathrm{fp}(\mathfrak{b}) = \big(d_b, \mathrm{labels}(\mathcal{O}_b), \mathrm{round}_{10}(\theta_b)\big)$. Among near-equivalent branches, keep the one with best lexicographic prune key $\pi_{\mathrm{prune}}(\mathfrak{b}) = \big(E_b, -S_b^{\mathrm{cum}}, K_b^{\mathrm{cum}}, |\mathcal{O}_b|, \mathrm{labels}(\mathcal{O}_b), \mathrm{round}_{10}(\theta_b), \mathrm{id}_b\big)$.

The beam operators are therefore $\mathrm{Dedup}(\mathcal{X}) = \{\text{best branch per fingerprint under } \pi_{\mathrm{prune}}\}$ and $\mathrm{Prune}(\mathcal{X}; B) = \mathrm{Top}_B^{\min}(\pi_{\mathrm{prune}}; \mathrm{Dedup}(\mathcal{X}))$. Hence the live and terminal pools update by $\mathfrak{F}_{c+1} = \mathrm{Prune}\Big(\bigcup_{b\in\mathfrak{F}_c} \mathfrak{C}_b^{\mathrm{front}}; B_{\mathrm{live}}\Big)$ and $\mathfrak{T}_{c+1} = \mathrm{Prune}\big(\mathfrak{T}_c \cup \bigcup_{b\in\mathfrak{F}_c} \mathfrak{C}_b^{\mathrm{term}}; B_{\mathrm{term}}\big)$.

(iv) Choose the final scaffold branch. After C scaffold-beam rounds, the surviving candidate set is $\mathfrak{W}_{\mathrm{final}} = \mathfrak{F}_C \cup \mathfrak{T}_C$, and the effective beam selector is $\mathfrak{b}_* = \arg\min_{\mathfrak{b}\in\mathfrak{W}_{\mathrm{final}}} \pi_{\mathrm{prune}}(\mathfrak{b})$.

A compact scaffold-beam algorithm block is therefore

$$
\begin{aligned}
&\text{A: } \mathfrak{F}_0 = \{\mathfrak{b}_{\mathrm{root}}\}, \qquad \mathfrak{T}_0 = \varnothing,\\
&\text{B: } \mathcal{R}_b^{\mathrm{raw}} \to \mathcal{S}_3(\mathfrak{b}) \to \mathcal{A}_b \qquad (\mathfrak{b} \in \mathfrak{F}_c),\\
&\text{C: } \mathcal{Q}(b) = \{\mathrm{stop}\} \cup \mathcal{A}_b, \qquad \mathfrak{C}_b = \{\mathcal{T}(\mathfrak{b}, q) : q \in \mathcal{Q}(b)\},\\
&\text{D: } \mathfrak{C}_b^{\mathrm{front}} = \{\mathfrak{b}' \in \mathfrak{C}_b : \mathrm{status}_{b'} = \mathrm{frontier}\}, \qquad \mathfrak{C}_b^{\mathrm{term}} = \{\mathfrak{b}' \in \mathfrak{C}_b : \mathrm{status}_{b'} = \mathrm{terminal}\},\\
&\text{E: } \mathfrak{F}_{c+1} = \mathrm{Prune}\left(\bigcup_{b\in\mathfrak{F}_c} \mathfrak{C}_b^{\mathrm{front}}; B_{\mathrm{live}}\right), \qquad \mathfrak{T}_{c+1} = \mathrm{Prune}\left(\mathfrak{T}_c \cup \bigcup_{b\in\mathfrak{F}_c} \mathfrak{C}_b^{\mathrm{term}}; B_{\mathrm{term}}\right),\\
&\text{F: } \mathfrak{W}_{\mathrm{final}} = \mathfrak{F}_C \cup \mathfrak{T}_C, \qquad \mathfrak{b}_* = \arg\min_{\mathfrak{b}\in\mathfrak{W}_{\mathrm{final}}} \pi_{\mathrm{prune}}(\mathfrak{b}).
\end{aligned}
$$

11.6 Active HH pool surface and selected scaffold

Identify the nested active HH pool with the retained generator images by $\Omega_{\mathrm{HH}}^{(k)} := \mathcal{G}^{(k)}$ for $k \in \{1, 2, 3\}$. The active HH adaptive surface is therefore carried by $\Omega_{\mathrm{HH}}^{(3)} \subseteq \Omega_{\mathrm{HH}}^{(2)} \subseteq \Omega_{\mathrm{HH}}^{(1)}$, with $\mathcal{G}_{\mathrm{adapt}}^{(k)} = \{\tau_m\}_{m\in\Omega_{\mathrm{HH}}^{(k)}}$.

If beam continuation is used, set $\mathcal{O}_* := \mathcal{O}_{\mathfrak{b}_*}$ and $\theta_*^{\mathrm{adapt}} := \theta_{\mathfrak{b}_*}$; otherwise $(\mathcal{O}_*, \theta_*^{\mathrm{adapt}})$ denotes the final admitted scaffold and its refit amplitudes on the main branch. The resulting state is $|\psi_*\rangle = U(\theta_*^{\mathrm{adapt}}; \mathcal{O}_*)|\phi_0\rangle$, so the active HH scaffold manifold selected by ADAPT may be written as $\mathcal{M}_{\mathrm{scaf}}(\mathcal{O}_*) = \{\, U(\vartheta; \mathcal{O}_*)|\phi_0\rangle : \vartheta \in \mathbb{R}^{|\mathcal{O}_*|} \,\}$.

17A. Projective McLachlan Real-Time Dynamics and Checkpoint-Adaptive Snake Control

This chapter is the real-time analogue of §11. The governing surface is the same staged-controller picture: fixed local geometry on the current scaffold, checkpoint-local structural search over candidate-position records $r = (m, p)$, scout/confirm/commit screening for append actions, and checkpoint-local scaffold pruning to keep the tangent plane small. The intended snake doctrine is:

$$\text{McLachlan} = \text{continuous within-scaffold motion law}, \qquad \text{controller} = \text{discrete append/prune law at checkpoints.}$$

Thus the scaffold is piecewise constant in physical time, and may change only at checkpoint events. Ordinary McLachlan dynamics is a local projection law on a fixed scaffold: at the current state it asks for the parameter-velocity vector whose tangent combination best matches the physical Schrödinger drift after removal of the physically irrelevant global-phase direction. Equivalently,

$$\dot{\theta}_\lambda \in \arg\min_{v \in \mathbb{R}^{n_\theta}} \Big\{ \|\bar{T}v - \bar{b}\|^2 + v^\top \Lambda v \Big\}.$$

Its practical outputs are the tangent metric $\bar{G}$, the force vector $\bar{f}$, and the projected residual $\bar{r}$; this residual is the correct local inadequacy signal for the current scaffold, and its natural normalization scale is the energy variance $\mathrm{Var}_\psi(H)$. Therefore McLachlan is treated here as the continuous within-scaffold motion oracle, while the checkpoint controller alone decides when the scaffold should expand or contract. The live controller-defining expressions are

$$\rho_{\mathrm{miss},k}, \qquad \Gamma_k^{\mathrm{live}}, \qquad \Sigma_{\mathrm{scout}}(r;k), \qquad \Delta_{\mathrm{McL}}(r;\tau_k), \qquad \Sigma_{\mathrm{conf}}(r;k), \qquad P_j^{\mathrm{pr}}(k),$$

and the displays below isolate these quantities in terms of the primitive projective objects Q_ψ, $\bar{b}$, $\bar{T}$, K, and $\bar{r}$. The older branch-level rollout and beam doctrine is retained only as a future extension, not as the canonical live runtime surface.

17A.1 Fixed-scaffold projective geometry

Fix a scaffold $\mathcal{O}$, state family $|\psi(\theta;\mathcal{O})\rangle$ with $\theta \in \mathbb{R}^{n_\theta}$, and Hamiltonian $H(\tau)$. At the current state, let $d := \dim \mathcal{H}$, let

$$Q_\psi := I - |\psi\rangle\langle\psi|, \qquad E_\psi := \langle\psi|H|\psi\rangle,$$

and define the projected Schrödinger drift vector by

$$\bar{b} := Q_\psi(-iH\psi) = \begin{bmatrix} (Q_\psi(-iH\psi))_1 \\ \vdots \\ (Q_\psi(-iH\psi))_d \end{bmatrix} = \begin{bmatrix} (-i(H - E_\psi)\psi)_1 \\ \vdots \\ (-i(H - E_\psi)\psi)_d \end{bmatrix} \in \mathbb{C}^d.$$

Define the projected tangent matrix by

$$\bar{T} := \begin{bmatrix} (Q_\psi\, \partial_{\theta_1}|\psi(\theta;\mathcal{O})\rangle)_1 & \cdots & (Q_\psi\, \partial_{\theta_{n_\theta}}|\psi(\theta;\mathcal{O})\rangle)_1 \\ \vdots & \ddots & \vdots \\ (Q_\psi\, \partial_{\theta_1}|\psi(\theta;\mathcal{O})\rangle)_d & \cdots & (Q_\psi\, \partial_{\theta_{n_\theta}}|\psi(\theta;\mathcal{O})\rangle)_d \end{bmatrix} \in \mathbb{C}^{d \times n_\theta}.$$

Define the regularizer by

$$\Lambda := \begin{bmatrix} \Lambda_{11} & \cdots & \Lambda_{1n_\theta} \\ \vdots & \ddots & \vdots \\ \Lambda_{n_\theta 1} & \cdots & \Lambda_{n_\theta n_\theta} \end{bmatrix} \in \mathbb{R}^{n_\theta \times n_\theta}, \qquad \Lambda^\top = \Lambda, \qquad \Lambda \succeq 0,$$

with simplest default $\Lambda = \lambda_{\text{ridge}} I$. Then the real McLachlan metric, force, and regularized system are

$$\bar{G} := \Re(\bar{T}^\dagger \bar{T}) = \begin{bmatrix} \Re(\sum_{a=1}^d (\bar{T})^*_{a1} (\bar{T})_{a1}) & \cdots & \Re(\sum_{a=1}^d (\bar{T})^*_{a1} (\bar{T})_{an_\theta}) \\ \vdots & \ddots & \vdots \\ \Re(\sum_{a=1}^d (\bar{T})^*_{an_\theta} (\bar{T})_{a1}) & \cdots & \Re(\sum_{a=1}^d (\bar{T})^*_{an_\theta} (\bar{T})_{an_\theta}) \end{bmatrix},$$

$$\bar{f} := \Re(\bar{T}^\dagger \bar{b}) = \begin{bmatrix} \Re(\sum_{a=1}^d (\bar{T})^*_{a1}\, \bar{b}_a) \\ \vdots \\ \Re(\sum_{a=1}^d (\bar{T})^*_{an_\theta}\, \bar{b}_a) \end{bmatrix}, \qquad K := \bar{G} + \Lambda, \qquad K\dot{\theta}_\lambda = \bar{f}.$$

Equivalently, on the fixed scaffold $\mathcal{O}$, ordinary McLachlan dynamics solves

$$\dot{\theta}_\lambda \in \arg\min_{v \in \mathbb{R}^{n_\theta}} \Big\{ \|\bar{T}v - \bar{b}\|^2 + v^\top \Lambda v \Big\},$$

so the only role of the current scaffold is to define the available tangent plane. The solve itself is therefore a within-scaffold motion law, not a scaffold-construction law. On a fixed scaffold $\mathcal{O}_k$, the runtime first solves $K\dot{\theta}_\lambda = \bar{f}$ and propagates $\theta(\tau)$ over the next checkpoint segment; it then measures $\rho_{\text{miss},k}$, and only if that miss is too large does the discrete checkpoint controller open the structural action set $\{\text{stay}, \text{append}(r), \text{prune}(j)\}$. The baseline miss telemetry is

$$\epsilon^2_{\text{proj}} := \|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f}, \qquad \epsilon^2_{\text{step}} := \|\bar{T}\dot{\theta}_\lambda - \bar{b}\|^2, \qquad \rho_{\text{miss}} := \frac{\epsilon^2_{\text{proj}}}{\|\bar{b}\|^2 + \varepsilon} = \frac{\epsilon^2_{\text{proj}}}{\text{Var}_\psi(H) + \varepsilon}.$$

So the miss statistic itself may be read as

$$\begin{aligned} \epsilon^2_{\text{proj}} &= \|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f} \\ &= \|Q_\psi(-iH\psi)\|^2 - \big(\Re(\bar{T}^\dagger \bar{b})\big)^\top \big(\Re(\bar{T}^\dagger \bar{T})\big)^+ \Re(\bar{T}^\dagger \bar{b}), \\ \rho_{\text{miss}} &= \frac{\epsilon^2_{\text{proj}}}{\text{Var}_\psi(H) + \varepsilon}. \end{aligned}$$

For any admissible checkpoint-local tangent baseline A with basis matrix $\bar{T}_{k,A}$ (for example a coordinate subset of the current scaffold tangents, or any chosen real tangent subspace with an explicit basis), define

$$\bar{G}_{k,A} := \Re(\bar{T}^\dagger_{k,A} \bar{T}_{k,A}), \qquad \bar{f}_{k,A} := \Re(\bar{T}^\dagger_{k,A} \bar{b}_k),$$

and the intrinsic reduced-space projective miss

$$\epsilon^2_{\text{proj},k}(A) := \|\bar{b}_k\|^2 - \bar{f}^\top_{k,A} \bar{G}^+_{k,A} \bar{f}_{k,A}, \qquad \rho_{\text{miss},k}(A) := \frac{\epsilon^2_{\text{proj},k}(A)}{\|\bar{b}_k\|^2 + \varepsilon}.$$

This quantity is exact for the reduced tangent space $\text{span}_{\mathbb{R}}(\bar{T}_{k,A})$. If $A \subseteq A'$ as tangent subspaces, then

$$\epsilon^2_{\text{proj},k}(A') \le \epsilon^2_{\text{proj},k}(A), \qquad \rho_{\text{miss},k}(A') \le \rho_{\text{miss},k}(A),$$

because enlarging the tangent baseline can only decrease the orthogonal projection distance of $\bar{b}_k$. In particular, if $A^{\text{sc}}_k \subseteq A^{\text{cm}}_k$, then

$$\rho_{\text{miss},k}(A^{\text{cm}}_k) \le \rho_{\text{miss},k}(A^{\text{sc}}_k),$$

so the scout-baseline miss is conservative for the larger shared commit baseline. The projective and raw residuals are

$$\bar{r} := \bar{T}\dot{\theta}_\lambda - \bar{b}, \qquad r_{\text{raw}} := T\dot{\theta}_\lambda + iH\psi, \qquad \bar{r} = Q_\psi r_{\text{raw}}.$$

Thus $\bar{r}$ is the local defect after the best available tangent velocity has been chosen on the current scaffold, and ρ_{miss} is the corresponding local inadequacy ratio relative to exact projective drift. So, for $\tau \in [\tau_k, \tau_{k+1})$, the live controller holds $\mathcal{O}(\tau) = \mathcal{O}_k$ fixed and uses McLachlan only to evolve $\theta(\tau)$ on that scaffold.

17A.2 Checkpoint miss test and telemetry-driven controller outputs

At checkpoint k, with physical time τ_k, the controller baseline summary is

$$\Xi_k^{\text{base}} := (E_k,\, \text{Var}_k,\, \epsilon^2_{\text{proj},k},\, \epsilon^2_{\text{step},k},\, \rho_{\text{miss},k},\, \|\dot{\theta}_{\lambda,k}\|_2,\, \text{rank}(K_k),\, \text{cond}(K_k)).$$

Let

$$A_k^{\text{sc}} \subseteq A_k^{\text{cm}} \subseteq \{1, \ldots, n_{\theta,k}\}$$

denote the scout and shared commit baselines at checkpoint k. Unless decorated otherwise, the live checkpoint shorthand is

$$\epsilon^2_{\text{proj},k} := \epsilon^2_{\text{proj},k}(A_k^{\text{sc}}), \qquad \rho_{\text{miss},k} := \rho_{\text{miss},k}(A_k^{\text{sc}}).$$

The clean rule is

$$\text{stay on } \mathcal{O}_k \text{ if } \rho_{\text{miss},k} \le \tau_{\text{miss}}, \qquad \text{open structural search otherwise.}$$

The `fullclone_3` runtime enriches this baseline with checkpoint cadence and telemetry-driven controller outputs. If $\dot{\theta}_k, \dot{\theta}_{k-1}, \dot{\theta}_{k-2}$ denote the aligned recent solved steps, define

$$\delta_k := \dot{\theta}_k - \dot{\theta}_{k-1}, \qquad \alpha_k := \delta_k - \delta_{k-1}, \qquad \rho_k^{\text{rate}} := \frac{\|\delta_k\|_2}{\max(\|\dot{\theta}_k\|_2, \|\dot{\theta}_{k-1}\|_2, \varepsilon)},$$

$$c_k^{\text{dir}} := \cos(\dot{\theta}_k, \dot{\theta}_{k-1}), \qquad c_k^{\text{curv}} := \cos(\alpha_k, \alpha_{k-1}), \qquad d_k^{\text{pred}} := \text{predicted displacement at checkpoint } k.$$

Then the controller uses the indicators

$$\mathbf{1}_k^{\text{rev}} := \mathbf{1}\big[c_k^{\text{dir}} \le \tau_{\text{rev}} \ \wedge\ \rho_k^{\text{rate}} \ge \tau_{\text{calm}}^{\text{rate}} \ \wedge\ \|\delta_k\|_2 \ge \tau_{\text{acc}}\big],$$

$$\mathbf{1}_k^{\text{flip}} := \mathbf{1}\big[c_k^{\text{curv}} \le \tau_{\text{flip}} \ \wedge\ \|\alpha_k\|_2 \ge \tau_{\text{acc}} \ \wedge\ \|\alpha_{k-1}\|_2 \ge \tau_{\text{acc}}\big],$$

$$\mathbf{1}_k^{\text{kink}} := \mathbf{1}\big[\mathbf{1}_k^{\text{rev}} = 1 \ \vee\ \mathbf{1}_k^{\text{flip}} = 1 \ \vee\ (\|\delta_k\|_2 \ge \tau_{\text{acc}} \ \wedge\ \rho_k^{\text{rate}} \ge \tau_{\text{kink}}^{\text{rate}})\big],$$

$$\mathbf{1}_k^{\text{calm}} := \mathbf{1}\big[k > 0 \ \wedge\ c_k^{\text{dir}} \ge \tau_{\text{calm}}^{\text{dir}} \ \wedge\ \rho_k^{\text{rate}} \le \tau_{\text{calm}}^{\text{rate}} \ \wedge\ \mathbf{1}_k^{\text{rev}} = 0 \ \wedge\ \mathbf{1}_k^{\text{flip}} = 0 \ \wedge\ d_k^{\text{pred}} \le 0.05\big].$$

If N_{short}^0 and f_{short}^0 are the base shortlist size and fraction, then

$$N_{\text{short}}(k) = \begin{cases} \max(1, N_{\text{short}}^0 + b_{\text{kink}}^{\text{short}}), & \mathbf{1}_k^{\text{kink}} = 1, \\ \max(1, \lceil s_{\text{calm}}^{\text{short}} N_{\text{short}}^0 \rceil), & \mathbf{1}_k^{\text{calm}} = 1, \\ N_{\text{short}}^0, & \text{otherwise}, \end{cases}$$

$$f_{\text{short}}(k) = \begin{cases} \min(1, 1.5\, f_{\text{short}}^0), & \mathbf{1}_k^{\text{kink}} = 1, \\ \max(0.05, s_{\text{calm}}^{\text{short}} f_{\text{short}}^0), & \mathbf{1}_k^{\text{calm}} = 1, \\ f_{\text{short}}^0, & \text{otherwise}, \end{cases}$$

while the effective refresh pressure is

$$\pi_k := \max_{\text{ord}}\big(\pi_k^{\text{base}}, \pi_k^{\text{tele}}\big), \qquad \pi_k^{\text{tele}} = \begin{cases} \text{medium}, & k = 0, \\ \text{high}, & \mathbf{1}_k^{\text{kink}} = 1, \\ \text{low}, & \text{otherwise.} \end{cases}$$

The oracle-budget multiplier is then

$$\beta_{\text{orc}}(k) = \begin{cases} \max\big(1, s_{\text{kink}}^{\text{orc}}\big), & \mathbf{1}_k^{\text{kink}} = 1 \ \vee \ \pi_k = \text{high}, \\ \max\big(0.25, s_{\text{calm}}^{\text{orc}}\big), & \mathbf{1}_k^{\text{calm}} = 1 \ \wedge \ \pi_k = \text{low}, \\ 1, & \text{otherwise.} \end{cases}$$

17A.3 Live checkpoint controller: stay, append, and prune

At each checkpoint, the discrete structural controller applies

$$\Gamma_k^{\text{live}} : (\mathcal{O}_k, \theta_k^-, \Xi_k^{\text{base}}) \longmapsto (\mathcal{O}_{k+1}, \theta_{k+1}^+),$$

with action set

$$\mathcal{U}_k^{\text{live}} := \{\text{stay}\} \cup \{\text{append}(r) : r \in \mathcal{R}_k^{\text{probe}}\} \cup \{\text{prune}(j) : j \in \mathcal{D}_k^{\text{prune}}\}.$$

Thus the live controller separates the continuous within-scaffold McLachlan motion from the discrete checkpoint-local scaffold edits. The append surface is the candidate-position record set from the current checkpoint, while the prune surface is the mature scaffold-coordinate set selected by the local prune controller. Only one discrete action is committed per checkpoint. The controller is lane-ordered rather than symmetric:

$$\text{lane}_k := \begin{cases} \text{append}, & \rho_{\text{miss},k} > \tau_{\text{miss}}, \\ \text{prune}, & \rho_{\text{miss},k} \le \tau_{\text{miss}} \ \wedge \ \Gamma_k^{\text{prune,perm}} = 1, \\ \text{stay}, & \text{otherwise.} \end{cases}$$

Hence a large miss opens the append lane, pruning is visible only in the calm low-miss regime, and failed append confirmation falls back to stay, not to prune at the same checkpoint.

17A.4 Append broad screen over candidate-position records

Candidate append is position-aware:

$$r = (m, p), \qquad \mathcal{O}_r^+ := \mathcal{O}_k \oplus_p m, \qquad \theta_r^+ := \theta_k \oplus_p 0.$$

The probed positions are restricted to a capped active set built from a fixed active suffix window,

$$W_k^{\text{act}} := \{\max(0, n_{\theta,k} - \omega_{\text{act}}), \ldots, n_{\theta,k} - 1\},$$

$$\mathcal{P}_k := \text{AllowedPositions}\big(n_{\theta,k}, \text{append} = n_{\theta,k}, W_k^{\text{act}}, N_{\text{probe}}^{\max}\big) \subseteq \{0, \ldots, n_{\theta,k}\}, \qquad |\mathcal{P}_k| \le N_{\text{probe}}^{\max}.$$

At zero initialization,

$$|\psi_r^+(\theta_k, 0)\rangle = |\psi_k\rangle, \qquad Q_{\psi_r^+} = Q_{\psi_k},$$

and the old tangent columns transport exactly. If $r = (m, p)$ inserts a q_r-parameter candidate block at position p, define the old-index transport

$$I_{r,p}(j) := \begin{cases} j, & j \le p, \\ j + q_r, & j > p. \end{cases}$$

Then

$$\bar{\tau}^{+}_{k,r,I_{r,p}(j)} = \bar{\tau}_{k,j},$$

hence for every old tangent baseline A,

$$\bar{G}^{+}_{k,I_{r,p}(A),I_{r,p}(A)} = \bar{G}_{k,A}, \qquad \bar{f}^{+}_{k,I_{r,p}(A)} = \bar{f}_{k,A}, \qquad K^{+}_{k,I_{r,p}(A),I_{r,p}(A)} = K_{k,A}.$$

Thus zero-init append transports the full old-old geometry exactly and requires no remeasurement of old-old blocks.

At checkpoint k, let $x_k \in \mathbb{R}^{M_k}$ collect the measurement bank already available on the current scaffold. Then every current old-old block is a fixed linear image of x_k:

$$\mathrm{vec}(\bar{G}_{k,A}) = L_{G,A} x_k, \qquad \bar{f}_{k,A} = L_{f,A} x_k, \qquad \|\bar{b}_k\|^2 = \ell_{b,k}^{\top} x_k.$$

For a candidate r, let

$$x^{\mathrm{aug}}_{k,r} := \begin{bmatrix} x_k \\ y_{k,r} \end{bmatrix}$$

augment the bank only by those candidate-local observables not already present. Then scout, confirm, and commit are nested linear images of this common candidate-local augmentation bank:

$$w^{\mathrm{cm}}_{r,k} = L^{\mathrm{sc}}_{w,r} x^{\mathrm{aug}}_{k,r}, \qquad \mathrm{vec}(B^{\mathrm{cm}}_{r,k}) = L^{\mathrm{cm}}_{B,r} x^{\mathrm{aug}}_{k,r}, \qquad \mathrm{vec}(C_{r,k}) = L_{C,r} x^{\mathrm{aug}}_{k,r}.$$

Scout therefore needs only $w^{\mathrm{cm}}_{r,k}$ and $C_{r,k}$; confirm requests only selected K^{cm}_k-whitened rows of $B^{\mathrm{cm}}_{r,k}$; commit may materialize the full old-new block.

Now fix the shared commit baseline A^{cm}_k. Write

$$\bar{T}^{\mathrm{cm}}_k := \bar{T}_{k,A^{\mathrm{cm}}_k}, \qquad \bar{G}^{\mathrm{cm}}_k := \bar{G}_{k,A^{\mathrm{cm}}_k}, \qquad \bar{f}^{\mathrm{cm}}_k := \bar{f}_{k,A^{\mathrm{cm}}_k}, \qquad K^{\mathrm{cm}}_k := \bar{G}^{\mathrm{cm}}_k + \Lambda^{\mathrm{cm}}_k,$$

and let

$$K^{\mathrm{cm}}_k \dot{\theta}^{\mathrm{cm}}_{k,\lambda} = \bar{f}^{\mathrm{cm}}_k, \qquad \bar{r}^{\mathrm{cm}}_k := \bar{T}^{\mathrm{cm}}_k \dot{\theta}^{\mathrm{cm}}_{k,\lambda} - \bar{b}_k.$$

The tangent space still gains a new column

$$u_r := \partial_{\vartheta_r} |\psi^{+}_r(\theta_k, \vartheta_r)\rangle\Big|_{\vartheta_r = 0}, \qquad \bar{u}_r := Q_\psi u_r.$$

For the candidate block $\bar{U}_r \in \mathbb{C}^{d \times q_r}$, define

$$B^{\mathrm{cm}}_{r,k} := \Re\big((\bar{T}^{\mathrm{cm}}_k)^{\dagger} \bar{U}_r\big), \qquad C_{r,k} := \Re(\bar{U}_r^{\dagger} \bar{U}_r), \qquad q_{r,k} := \Re(\bar{U}_r^{\dagger} \bar{b}_k),$$

and the residualized force on the shared commit baseline

$$w^{\mathrm{cm}}_{r,k} := q_{r,k} - (B^{\mathrm{cm}}_{r,k})^{\top} (K^{\mathrm{cm}}_k)^{-1} \bar{f}^{\mathrm{cm}}_k = -\Re(\bar{U}_r^{\dagger} \bar{r}^{\mathrm{cm}}_k).$$

The scout-safe lower-bound Schur gain is

$$\underline{\Delta}_{\mathrm{McL}}(r; \tau_k) := (w^{\mathrm{cm}}_{r,k})^{\top} (C_{r,k} + \Lambda_r)^{+} w^{\mathrm{cm}}_{r,k} = \max_{\eta \in \mathbb{R}^{q_r}} \left\{ 2\eta^{\top} w^{\mathrm{cm}}_{r,k} - \eta^{\top} (C_{r,k} + \Lambda_r) \eta \right\}.$$

This is a conservative lower bound to the exact commit-grade gain because

$$S_{r,k}^{\mathrm{cm}} := C_{r,k} + \Lambda_r - (B_{r,k}^{\mathrm{cm}})^\top (K_k^{\mathrm{cm}})^{-1} B_{r,k}^{\mathrm{cm}} \preceq C_{r,k} + \Lambda_r.$$

The normalized scout score is therefore

$$\underline{\rho}_{\mathrm{gain}}(r;k) := \frac{\underline{\Delta}_{\mathrm{McL}}(r;\tau_k)}{\|\bar{b}_k\|^2 + \varepsilon},$$

$$\Sigma_{\mathrm{scout}}(r;k) := \underline{\rho}_{\mathrm{gain}}(r;k) + b_{\mathrm{temp}}(r;k) - \beta_{\mathrm{comp}}\, C_{\mathrm{comp}}(r;k) - \beta_{\mathrm{meas}}\, G_{\mathrm{new}}(r;k) - \beta_{\mathrm{jump}}\, J_{\mathrm{jump}}(r;k).$$

Hence scout is already written on the same McLachlan geometry as commit, but uses only the residualized force and the candidate self-block rather than the full old-new Schur data.

17A.5 Append confirm, compressed rerank, and exact commit

The runtime remains organized in three tiers,

$$\text{scout}, \qquad \text{confirm}, \qquad \text{commit},$$

but confirm now enriches the same candidate on the shared commit baseline by a monotone family of K_k^{cm}-whitened old-mode responses.

Let

$$K_k^{\mathrm{cm}} = V_k \Sigma_k^2 V_k^\top$$

be the spectral factorization of the shared commit baseline on its numerical support, and define the corresponding K_k^{cm}-whitened coordinates

$$z_k := \Sigma_k^{-1} V_k^\top \bar{f}_k^{\mathrm{cm}}, \qquad \Gamma_{r,k} := \Sigma_k^{-1} V_k^\top B_{r,k}^{\mathrm{cm}}.$$

Then

$$w_{r,k}^{\mathrm{cm}} = q_{r,k} - \Gamma_{r,k}^\top z_k, \qquad S_{r,k}^{\mathrm{cm}} = C_{r,k} + \Lambda_r - \Gamma_{r,k}^\top \Gamma_{r,k}.$$

Let

$$J_r^{(0)} \subseteq J_r^{(1)} \subseteq \cdots \subseteq J_r^{(m_{\max})} \subseteq \{1, \ldots, \mathrm{rank}(K_k^{\mathrm{cm}})\}$$

be any nested candidate-local family of whitened baseline mode sets, and let $\Pi_{J_r^{(m)}}$ denote the coordinate projector onto $J_r^{(m)}$. A deterministic implementation may, for example, order the modes by decreasing row norm of $\Gamma_{r,k}$, but the ladder below does not depend on the chosen ordering. Define the compressed Schur family

$$\widetilde{S}_{r,k}^{(m)} := C_{r,k} + \Lambda_r - \Gamma_{r,k}^\top \Pi_{J_r^{(m)}} \Gamma_{r,k},$$

and the corresponding confirm-grade lower-bound gains

$$\widetilde{\Delta}_{\mathrm{McL}}^{(m)}(r;\tau_k, J_r) := (w_{r,k}^{\mathrm{cm}})^\top (\widetilde{S}_{r,k}^{(m)})^+ w_{r,k}^{\mathrm{cm}}.$$

Equivalently,

$$\widetilde{\Delta}_{\mathrm{McL}}^{(m)}(r;\tau_k, J_r) = \max_{\eta \in \mathbb{R}^{q_r},\, \zeta \in \mathrm{range}(\Pi_{J_r^{(m)}})} \left\{ 2\eta^\top w_{r,k}^{\mathrm{cm}} - \eta^\top (C_{r,k} + \Lambda_r)\eta - 2\zeta^\top \Gamma_{r,k}\eta - \|\zeta\|_2^2 \right\}.$$

Thus the confirm ladder is monotone:

$$\underline{\Delta}_{\mathrm{McL}}(r;\tau_k) = \widetilde{\Delta}_{\mathrm{McL}}^{(0)}(r;\tau_k, J_r) \le \widetilde{\Delta}_{\mathrm{McL}}^{(1)}(r;\tau_k, J_r) \le \cdots \le \widetilde{\Delta}_{\mathrm{McL}}^{(m_{\max})}(r;\tau_k, J_r) \le \Delta_{\mathrm{McL}}(r;\tau_k, A_k^{\mathrm{cm}}),$$

where the exact commit-grade gain on the shared baseline is

$$\Delta_{\mathrm{McL}}(r;\tau_k,A_k^{\mathrm{cm}}) := (w_{r,k}^{\mathrm{cm}})^\top (S_{r,k}^{\mathrm{cm}})^+ w_{r,k}^{\mathrm{cm}}.$$

The normalized confirm and commit gains are

$$\widetilde{\rho}_{\mathrm{gain}}^{(m)}(r;k) := \frac{\widetilde{\Delta}_{\mathrm{McL}}^{(m)}(r;\tau_k,J_r)}{\|\bar{b}_k\|^2+\varepsilon}, \qquad \rho_{\mathrm{gain}}^{\mathrm{cm}}(r;k) := \frac{\Delta_{\mathrm{McL}}(r;\tau_k,A_k^{\mathrm{cm}})}{\|\bar{b}_k\|^2+\varepsilon}.$$

Scout keeps

$$\mathcal{S}_k^{\mathrm{scout}} := \mathrm{Top}_{N_{\mathrm{short}}(k)}\big(\Sigma_{\mathrm{scout}}(r;k);\ r\in\mathcal{R}_k^{\mathrm{probe}}\big).$$

Confirm evaluates one chosen compressed level m_r for each scout survivor and ranks with

$$\Sigma_{\mathrm{conf}}(r;k) := \widetilde{\rho}_{\mathrm{gain}}^{(m_r)}(r;k) - \beta_{\mathrm{meas}}\, G_{\mathrm{new}}(r;k) - \beta_{\mathrm{dir}}\, D_{\mathrm{dir}}(r;k).$$

The confirm shortlist is therefore

$$\mathcal{S}_k^{\mathrm{conf}} := \mathrm{Top}_{\max\{1,\lceil f_{\mathrm{short}}(k)\,|\mathcal{S}_k^{\mathrm{scout}}|\rceil\}}\big(\Sigma_{\mathrm{conf}}(r;k);\ r\in\mathcal{S}_k^{\mathrm{scout}}\big).$$

The selected append record is

$$r_k^\star := \arg\max_{r\in\mathcal{S}_k^{\mathrm{conf}}} \Sigma_{\mathrm{conf}}(r;k).$$

The clean commit rule is

$$\text{append } r_k^\star \iff \rho_{\mathrm{miss},k}(A_k^{\mathrm{cm}}) > \tau_{\mathrm{miss}}, \quad \rho_{\mathrm{gain}}^{\mathrm{cm}}(r_k^\star;k) \ge \tau_{\mathrm{gain}}, \quad \Delta_{\mathrm{McL}}(r_k^\star;\tau_k,A_k^{\mathrm{cm}}) \ge \delta_{\mathrm{append}}.$$

Otherwise the controller stays on $\mathcal{O}_k$. Thus scout, confirm, and commit all remain inside the same McLachlan geometry, but use a staged Schur family that narrows measurement effort from residualized force, to selected old-mode responses, to the exact shared-baseline Schur complement.

17A.6 Local scaffold pruning and rollback

Pruning acts on scaffold coordinates, not on branch objects. Let the current scaffold be $\mathcal{O}_k = (m_1,\dots,m_{n_k})$ with $n_k := |\mathcal{O}_k|$, and let each coordinate carry the metadata (ι_j,κ_j,c_j), where ι_j is the checkpoint at which coordinate j was appended, κ_j is its stored append-time burden, and $c_j(k)\in\{0,1,\dots,c_{\mathrm{cool}}\}$ is its prune-cooldown counter. The hard protection set is

$$\mathcal{P}_{\mathrm{protect}}(k) := \{\, j :\ k-\iota_j < \ell_{\mathrm{protect}} \,\},$$

so coordinates inside $\mathcal{P}_{\mathrm{protect}}(k)$ are never considered for pruning. For the recent-history stagnation window, let

$$\mathcal{I}_j^{\mathrm{stag}}(k) := \{\, \ell : \max(\iota_j+1, k-L_{\mathrm{stag}}+1) \le \ell \le k \,\}, \qquad \Delta\theta_{j,\ell} := \theta_j(\tau_\ell^-) - \theta_j(\tau_{\ell-1}^+), \qquad \bar{f}_{j,\ell} := (\bar{f}_\ell)_j.$$

Then the normalized recent-motion and fit-contribution summaries are

$$m_j^{\mathrm{mov}}(k) := \frac{1}{|\mathcal{I}_j^{\mathrm{stag}}(k)|} \sum_{\ell\in\mathcal{I}_j^{\mathrm{stag}}(k)} \frac{|\Delta\theta_{j,\ell}|}{\|\Delta\theta_\ell\|_\infty+\varepsilon_\theta},$$

$$m_j^{\mathrm{fit}}(k) := \frac{1}{|\mathcal{I}_j^{\mathrm{stag}}(k)|} \sum_{\ell\in\mathcal{I}_j^{\mathrm{stag}}(k)} \frac{|\Delta\theta_{j,\ell}|\,|\bar{f}_{j,\ell}|}{\sum_i |\Delta\theta_{i,\ell}|\,|\bar{f}_{i,\ell}|+\varepsilon_f}.$$

The hybrid checkpoint-local stagnation score is

$$s_j^{\mathrm{stag}}(k) := \alpha_{\mathrm{stag}}\Big(1 - \mathrm{clip}(m_j^{\mathrm{mov}}(k), 0, 1)\Big) + (1 - \alpha_{\mathrm{stag}})\Big(1 - \mathrm{clip}(m_j^{\mathrm{fit}}(k), 0, 1)\Big), \qquad 0 \le \alpha_{\mathrm{stag}} \le 1,$$

so larger values mean less recent motion and less recent contribution to local drift tracking. For coordinates outside the hard protection window, the optional soft early-removal penalty is

$$\Pi_j^{\mathrm{age}}(k) := 1 + \lambda_{\mathrm{age}}^{\mathrm{pr}} \exp\left(-\frac{k - \iota_j - \ell_{\mathrm{protect}}}{\ell_{\mathrm{age}}}\right), \qquad j \notin \mathcal{P}_{\mathrm{protect}}(k).$$

The mature eligible set is

$$\mathcal{E}_k^{\mathrm{prune}} := \{\, j : \ j \notin \mathcal{P}_{\mathrm{protect}}(k),\ c_j(k) = 0,\ |\mathcal{I}_j^{\mathrm{stag}}(k)| \ge 1,\ s_j^{\mathrm{stag}}(k) \ge s_{\mathrm{stale}} \,\}.$$

Pruning is opened only when the current scaffold is already tracking the drift adequately. The permissibility gate is

$$\Gamma_k^{\mathrm{prune,perm}} := \mathbf{1}[\rho_{\mathrm{miss},k} \le \tau_{\mathrm{prune}}^{\mathrm{miss}}]\, \mathbf{1}[c_k^{\mathrm{dir}} \ge \tau_{\mathrm{calm}}^{\mathrm{dir}}]\, \mathbf{1}[\rho_k^{\mathrm{rate}} \le \tau_{\mathrm{calm}}^{\mathrm{rate}}]\, \mathbf{1}[n_k \ge 2], \qquad \tau_{\mathrm{prune}}^{\mathrm{miss}} \le \tau_{\mathrm{miss}}.$$

Once the current baseline geometry $(\bar{G}_k^{\mathrm{cm}}, \bar{f}_k^{\mathrm{cm}})$ is cached, broad prune nomination is shot-free. For each $j \in \mathcal{E}_k^{\mathrm{prune}} \cap A_k^{\mathrm{cm}}$, let

$$A_{k,-j}^{\mathrm{cm}} := A_k^{\mathrm{cm}} \setminus \{j\}, \qquad \bar{G}_{k,-j} := \bar{G}_{k,A_{k,-j}^{\mathrm{cm}}}, \qquad \bar{f}_{k,-j} := \bar{f}_{k,A_{k,-j}^{\mathrm{cm}}}.$$

Define the cached frozen-ablation miss increase

$$L_{j,\mathrm{frz}}^{\mathrm{cache}}(k) := \left[\frac{\epsilon_{\mathrm{proj},k}^2(A_{k,-j}^{\mathrm{cm}}) - \epsilon_{\mathrm{proj},k}^2(A_k^{\mathrm{cm}})}{\|\bar{b}_k\|^2 + \varepsilon}\right]_+ = \left[\frac{(\bar{f}_k^{\mathrm{cm}})^\top (\bar{G}_k^{\mathrm{cm}})^+ \bar{f}_k^{\mathrm{cm}} - \bar{f}_{k,-j}^\top \bar{G}_{k,-j}^+ \bar{f}_{k,-j}}{\|\bar{b}_k\|^2 + \varepsilon}\right]_+.$$

This quantity depends only on cached submatrices and subvectors of the current geometry and therefore requires no new shots. The local prune probe set is then

$$\mathcal{D}_k^{\mathrm{prune}} := \begin{cases} \mathrm{Top}_{K_{\mathrm{pr}}}\Big(-L_{j,\mathrm{frz}}^{\mathrm{cache}}(k);\ j \in \mathcal{E}_k^{\mathrm{prune}} \cap A_k^{\mathrm{cm}}\Big), & \Gamma_k^{\mathrm{prune,perm}} = 1, \\ \varnothing, & \Gamma_k^{\mathrm{prune,perm}} = 0. \end{cases}$$

When cached losses tie numerically, ties are broken in favor of larger stored burden κ_j. Inside the nominated set, use the cheap shot-free prune score

$$P_j^{\mathrm{pr}}(k) := L_{j,\mathrm{frz}}^{\mathrm{cache}}(k), \qquad j_k^\star := \arg\min_{j \in \mathcal{D}_k^{\mathrm{prune}}} P_j^{\mathrm{pr}}(k).$$

Thus broad prune ranking is geometry-exact on the cached baseline and requires no new measurements; the later remove-and-reintegrate test remains the authoritative prune accept/reject stage. The short local prune horizon is fixed as

$$\Delta\tau_{\mathrm{pr}} := \eta_{\mathrm{pr}}(\tau_{k+1} - \tau_k), \qquad 0 < \eta_{\mathrm{pr}} \le 1.$$

The authoritative prune test removes $j_k^\star$, resolves McLachlan on the reduced scaffold over the short local probe segment $[\tau_k, \tau_k + \Delta\tau_{\mathrm{pr}}]$, and compares the reduced run against the incumbent by

$$J_{\mathrm{seg},k} := \int_{\tau_k}^{\tau_k + \Delta\tau_{\mathrm{pr}}} \|\bar{r}(\tau)\|^2 \, d\tau,$$

$$\Delta\rho_j^{\mathrm{pr}}(k) := \rho_{\mathrm{miss},k}^{(-j)} - \rho_{\mathrm{miss},k}, \qquad \Delta J_j^{\mathrm{pr}}(k) := \int_{\tau_k}^{\tau_k+\Delta\tau_{\mathrm{pr}}} \Big(\|\bar{r}^{(-j)}(\tau)\|^2 - \|\bar{r}(\tau)\|^2\Big)\, d\tau.$$

Accept the prune iff

$$\Delta\rho_{j_k^\star}^{\mathrm{pr}}(k) \le \Delta_{\mathrm{safe}}^{\mathrm{miss}}, \qquad \Delta J_{j_k^\star}^{\mathrm{pr}}(k) \le \Delta_{\mathrm{safe}}^{\mathrm{seg}}.$$

Otherwise rollback to the pre-prune scaffold and set $c_{j_k^\star}(k^+) \leftarrow c_{\mathrm{cool}}$. Thus the live checkpoint action is

$$\Gamma_k^{\mathrm{live}} \in \{\mathrm{stay}, \mathrm{append}(r_k^\star), \mathrm{prune}(j_k^\star)\},$$

where append is selected by the shortlist/confirm/commit controller above, and prune is selected by the local remove-and-reintegrate controller above. Between checkpoints, the scaffold is fixed; at checkpoints, only one discrete append or prune action is committed.

17A.7 Future branch-level rollout and beam extension

If one later lifts the local stay/append/prune controller to a branch controller, a natural future extension is to compare short rollout alternatives by segment residual cost. On a fixed scaffold over a segment $[\tau_a, \tau_b]$, the raw and projective segment costs are

$$J_{\mathrm{seg}}^{\mathrm{raw}} := \int_{\tau_a}^{\tau_b} \|r_{\mathrm{raw}}(\tau)\|^2\, d\tau, \qquad J_{\mathrm{seg}}^{\mathrm{proj}} := \int_{\tau_a}^{\tau_b} \|\bar{r}(\tau)\|^2\, d\tau.$$

A clean scalar branch objective is

$$J_{\mathrm{dyn}}(\mathfrak{b}) := \sum_{k=0}^{K-1} J_{\mathrm{seg},k} + \beta\, C_{\mathrm{hw}}(\mathfrak{b}) + \gamma\, N_{\mathrm{adapt}}(\mathfrak{b}).$$

The associated residual ledgers may be summarized by

$$\Delta J^{\epsilon} := \int \epsilon_{\mathrm{step}}^2, \qquad \Delta J^{\mathrm{model}} := \int \epsilon_{\mathrm{proj}}^2, \qquad \Delta J^{\mathrm{damp}} := \int \max\!\big(\epsilon_{\mathrm{step}}^2 - \epsilon_{\mathrm{proj}}^2, 0\big),$$

then culls rollout alternatives lexicographically after deduplicating by checkpoint index, scaffold labels, and rounded parameters. Thus the branch objective may be read as the integrated residual ledger plus explicit hardware and adaptation penalties:

$$\begin{aligned}
J_{\mathrm{dyn}}(\mathfrak{b}) &= \sum_{k=0}^{K-1} J_{\mathrm{seg},k} + \beta\, C_{\mathrm{hw}}(\mathfrak{b}) + \gamma\, N_{\mathrm{adapt}}(\mathfrak{b}),\\
J_{\mathrm{seg},k} &\sim \int_{\tau_k}^{\tau_{k+1}} \|\bar{r}(\tau)\|^2\, d\tau,\\
\Delta J^{\epsilon} &= \int \|\bar{T}\dot{\theta}_\lambda - \bar{b}\|^2,\\
\Delta J^{\mathrm{model}} &= \int \big(\|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f}\big),\\
\Delta J^{\mathrm{damp}} &= \int \max\!\big(\epsilon_{\mathrm{step}}^2 - \epsilon_{\mathrm{proj}}^2, 0\big).
\end{aligned}$$

This branch-level rollout and beam surface is retained only as a future extension. It is not the canonical live runtime surface of `fullclone_3`.

Implementation status. The repo surface in `fullclone_3` is already more structured than the original long-form derivation. It already implements checkpoint-local geometry caches, three measurement tiers, active-window and probe-position caps, position-aware zero-initialized single-parameter growth, and compile/measurement/directional/temporal-prior controller modifiers. The condensed doctrine is therefore:

Propagate on the current scaffold by projective McLachlan. Measure miss at each checkpoint. If the miss is large, open a position-aware append search over records $r = (m, p)$. Scout by residual overlap and hardware penalties, confirm by local Schur-complement gain, and append only if the miss and gain thresholds are passed. If the scaffold is already calm and well fit, prune stale scaffold generators by a local remove-and-reintegrate test with rollback. Treat branch-level rollout and beam competition only as a future extension.

17B. Projective McLachlan Real-Time Geometry and Piecewise Beam-Adaptive Control

Below is the cleanest mathematical story for what the projected real-time implementation is doing. It is most naturally written as a **piecewise-smooth control problem on a family of variational manifolds**, with $\hbar = 1$, normalized pure states, and real parameters. The two geometric facts underneath everything are:

1. the McLachlan principle projects the exact Schrödinger velocity onto a **real-linear** tangent space using the real inner product $g(u, v) = \Re\langle u, v\rangle$, and
2. physical pure states are rays, so the natural state space is **projective Hilbert space**, which means global-phase directions should be removed from the local scoring geometry.

17B.1 What problem is being solved?

Fix a Hilbert space $\mathcal{H} \cong \mathbb{C}^d$, a Hamiltonian $H(\tau)$, and a current ansatz scaffold $\mathcal{O}$. That scaffold defines a parameterized family of states

$$|\psi(\theta; \mathcal{O})\rangle, \qquad \theta \in \mathbb{R}^{n_\theta}.$$

At physical time τ, the exact Schrödinger equation asks for the state velocity

$$\frac{d}{d\tau}|\psi_{\text{ex}}(\tau)\rangle = -iH(\tau)|\psi_{\text{ex}}(\tau)\rangle.$$

The local projected-dynamics question is therefore:

> **At the current state, how closely can the current ansatz manifold reproduce the exact instantaneous quantum velocity?**

This is the local question answered by real-time McLachlan dynamics. It is also the quantity that drives adaptive manifold growth: enlarge the ansatz when the instantaneous projected defect becomes too large.

17B.2 Why projective Hilbert space is the right setting

A normalized ket $|\psi\rangle$ and $e^{i\phi}|\psi\rangle$ represent the same physical state. So the physical state space is projective Hilbert space $\mathbb{P}(\mathcal{H})$ rather than the unit sphere itself. At a normalized state $|\psi\rangle$, define the horizontal projector

$$Q_\psi := I - |\psi\rangle\langle\psi|.$$

For each parameter θ_j, the raw tangent vector is

$$\tau_j := \partial_{\theta_j}|\psi(\theta; \mathcal{O})\rangle,$$

and the horizontal tangent is

$$\bar{\tau}_j := Q_\psi \tau_j.$$

Collect the horizontal tangents as columns:

$$\bar{T} := [\bar{\tau}_1, \ldots, \bar{\tau}_{n_\theta}] \in \mathbb{C}^{d \times n_\theta}.$$

The exact Schrödinger drift is

$$b := -iH\psi,$$

but the physically relevant local velocity is its horizontal part

$$\bar{b} := Q_\psi b = -i(H - E_\psi)\psi, \qquad E_\psi := \langle\psi|H|\psi\rangle.$$

Its squared norm is

$$\|\bar{b}\|^2 = \langle\psi|(H - E_\psi)^2|\psi\rangle = \mathrm{Var}_\psi(H).$$

So the squared norm of the exact projective Schrödinger velocity is the energy variance. This is why $\mathrm{Var}_\psi(H)$ is the natural normalization scale for projected McLachlan defects.

17B.3 The current manifold and its allowed velocities

Because the parameters are real, the attainable instantaneous velocities form the **real-linear** tangent cone

$$T_\psi\mathcal{M}_\mathcal{O} = \left\{\bar{T}\dot{\theta} : \dot{\theta} \in \mathbb{R}^{n_\theta}\right\}.$$

This is not a complex-linear subspace in general. So the correct local geometry is built from the real inner product

$$g(u, v) := \Re\langle u, v\rangle.$$

The variational velocity associated with parameter rates $\dot{\theta}$ is

$$v(\dot{\theta}) = \bar{T}\dot{\theta}.$$

17B.4 The fixed-structure McLachlan step

The fixed-structure local problem is

$$\dot{\theta}_0 = \arg\min_{\dot{\theta}\in\mathbb{R}^{n_\theta}} \|\bar{T}\dot{\theta} - \bar{b}\|^2.$$

Its quadratic form, projective metric, force vector, and solved stationarity system close to

$$\begin{aligned}
\|\bar{T}\dot{\theta} - \bar{b}\|^2 &= \|\bar{b}\|^2 - 2\Re\langle\bar{T}\dot{\theta}, \bar{b}\rangle + \dot{\theta}^\top\Re(\bar{T}^\dagger\bar{T})\dot{\theta},\\
(\bar{G})_{ij} &= \Re\langle\bar{\tau}_i, \bar{\tau}_j\rangle = \Re\langle\partial_{\theta_i}\psi|Q_\psi|\partial_{\theta_j}\psi\rangle,\\
\bar{f}_i &= \Re\langle\bar{\tau}_i, \bar{b}\rangle = \Re\langle Q_\psi\partial_{\theta_i}\psi, -i(H - E_\psi)|\psi\rangle,\\
\mathcal{L}_0(\dot{\theta}) &= \|\bar{b}\|^2 - 2\bar{f}^\top\dot{\theta} + \dot{\theta}^\top\bar{G}\dot{\theta},\\
\bar{G}\dot{\theta}_0 &= \bar{f},\\
\dot{\theta}_0 &= \bar{G}^+\bar{f} \qquad \text{if } \bar{G} \text{ is singular or rank-deficient},\\
0 &= \Re\langle\bar{\tau}_i, \bar{T}\dot{\theta}_0 - \bar{b}\rangle \qquad \forall i.
\end{aligned}$$

17B.5 What regularization means

A stabilized runtime step solves

$$\dot{\theta}_\lambda = \arg\min_{\dot{\theta}\in\mathbb{R}^{n_\theta}} \left(\|\bar{T}\dot{\theta} - \bar{b}\|^2 + \dot{\theta}^\top\Lambda\dot{\theta}\right), \qquad \Lambda \succeq 0.$$

With

$$K := \bar{G} + \Lambda,$$

the regularized geometry closes to

$$
\begin{aligned}
&K\dot{\theta}_\lambda = \bar{f},\\
&\epsilon^2_{\text{proj}} := \min_{\dot{\theta}} \|\bar{T}\dot{\theta} - \bar{b}\|^2 = \|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f},\\
&\epsilon^2_{\text{step}} := \|\bar{T}\dot{\theta}_\lambda - \bar{b}\|^2,\\
&\mathcal{L}^\star_\lambda := \min_{\dot{\theta}} \left(\|\bar{T}\dot{\theta} - \bar{b}\|^2 + \dot{\theta}^\top \Lambda \dot{\theta}\right) = \|\bar{b}\|^2 - \bar{f}^\top K^{-1} \bar{f} \qquad (K \text{ invertible}),\\
&\mathcal{L}^\star_\lambda = \epsilon^2_{\text{step}} + \dot{\theta}_\lambda^\top \Lambda \dot{\theta}_\lambda.
\end{aligned}
$$

So ϵ^2_{proj} measures intrinsic model inadequacy, ϵ^2_{step} measures the actual mismatch of the damped runtime step, and $\mathcal{L}^\star_\lambda$ is the stabilized solver objective.

17B.6 Why the residual is the right local error quantity

Choose a local gauge in which the representatives are horizontal at the current physical time τ. Then the exact and variational projective velocities are

$$\bar{b} \qquad \text{and} \qquad \bar{T}\dot{\theta}.$$

Define the horizontal residual

$$\bar{r} := \bar{T}\dot{\theta} - \bar{b}.$$

Then, for a short step h,

$$|\psi_{\text{ex}}(\tau + h)\rangle = |\psi\rangle + h\,\bar{b} + O(h^2),$$

$$|\psi_{\text{var}}(\tau + h)\rangle = |\psi\rangle + h\,\bar{T}\dot{\theta} + O(h^2),$$

so

$$|\psi_{\text{var}}(\tau + h)\rangle - |\psi_{\text{ex}}(\tau + h)\rangle = h\,\bar{r} + O(h^2).$$

So the correct local score is an **instantaneous derivative-mismatch score**. Static energy scores or novelty scores can help in secondary ranking, but they are not the primary real-time control quantity.

17B.7 The geometric projector behind the formulas

Because the tangent space is real-linear, the correct projector is

$$P^{\mathbb{R}}_{\bar{T}} v = \bar{T}\,\bar{G}^+\,\Re(\bar{T}^\dagger v).$$

The unregularized optimal residual is therefore

$$\bar{r}_0 = \bar{T}\dot{\theta}_0 - \bar{b} = \left(P^{\mathbb{R}}_{\bar{T}} - I\right)\bar{b}.$$

So the ansatz follows the component of the exact projective Schrödinger vector field that lies in the current real tangent plane, and the residual is the orthogonal complement.

17B.8 What a candidate generator really is

Suppose the current scaffold is an ordered product ansatz

$$|\psi(\theta;\mathcal{O})\rangle = e^{-i\theta_{n_\theta}a_{n_\theta}}\cdots e^{-i\theta_1 a_1}|\phi_0\rangle.$$

Now fix a candidate-position record $r=(m,p)$, insert the candidate generator A_m at position p, and let the new parameter ϑ_r be initialized at zero:

$$|\psi_r^+(\theta,\vartheta_r)\rangle = e^{-i\theta_{n_\theta}a_{n_\theta}}\cdots e^{-i\theta_p a_p}e^{-i\vartheta_r A_m}e^{-i\theta_{p-1}a_{p-1}}\cdots e^{-i\theta_1 a_1}|\phi_0\rangle.$$

At $\vartheta_r=0$,

$$|\psi_r^+(\theta,0)\rangle = |\psi(\theta)\rangle.$$

So the state is continuous, but the tangent space changes immediately. The new tangent column is

$$u_r := \partial_{\vartheta_r}|\psi_r^+(\theta,\vartheta_r)\rangle\Big|_{\vartheta_r=0}, \qquad \bar{u}_r := Q_\psi u_r.$$

So a candidate is not merely "a promising operator." It is a **new tangent direction that becomes available at the current state without moving the state**. For a block candidate carrying q new real parameters, collect the new columns into

$$\bar{U}_r \in \mathbb{C}^{d\times q}.$$

17B.9 The exact local gain formula

Now augment the local problem:

$$\min_{\dot\theta,\eta}\Big(\|\bar{T}\dot\theta + \bar{U}_r\eta - \bar{b}\|^2 + \dot\theta^\top\Lambda\dot\theta + \eta^\top\Lambda_r\eta\Big), \qquad \Lambda_r \succeq 0.$$

With the mixed overlaps

$$\begin{aligned}
B_{r,ia} &= \Re\langle\bar{\tau}_i,\bar{u}_{r,a}\rangle, & B_r &= \Re(\bar{T}^\dagger\bar{U}_r)\in\mathbb{R}^{n_\theta\times q},\\
C_{r,ab} &= \Re\langle\bar{u}_{r,a},\bar{u}_{r,b}\rangle, & C_r &= \Re(\bar{U}_r^\dagger\bar{U}_r)\in\mathbb{R}^{q\times q},\\
q_{r,a} &= \Re\langle\bar{u}_{r,a},\bar{b}\rangle, & q_r &= \Re(\bar{U}_r^\dagger\bar{b})\in\mathbb{R}^{q},
\end{aligned}$$

the augmented normal equations, Schur complement, and gain reduce to

$$\begin{aligned}
\begin{bmatrix} K & B_r \\ B_r^\top & C_r+\Lambda_r\end{bmatrix}\begin{bmatrix}\dot\theta\\ \eta\end{bmatrix} &= \begin{bmatrix}\bar{f}\\ q_r\end{bmatrix},\\
\dot\theta(\eta) &= K^{-1}(\bar{f}-B_r\eta),\\
\mathcal{L}^\star_{\mathrm{aug},\lambda}(\eta) &= \mathcal{L}^\star_\lambda - 2\eta^\top w_r + \eta^\top S_r\eta,\\
S_r &= C_r+\Lambda_r - B_r^\top K^{-1}B_r,\\
w_r &= q_r - B_r^\top K^{-1}\bar{f},\\
\eta_\star &= S_r^+ w_r,\\
\Delta_{\mathrm{McL}}(r;\tau) &:= \mathcal{L}^\star_\lambda - \mathcal{L}^\star_{\mathrm{aug},\lambda} = w_r^\top S_r^+ w_r\\
&= (q_r - B_r^\top K^{-1}\bar{f})^\top (C_r+\Lambda_r - B_r^\top K^{-1}B_r)^+(q_r - B_r^\top K^{-1}\bar{f}).
\end{aligned}$$

For a single real candidate direction ($q=1$), this reduces to

$$\Delta_{\mathrm{McL}}(r;\tau) = \frac{w_r^2}{S_r}.$$

17B.10 What S and w mean

The Schur-complement pair and resulting score may be read in one chain:

$$\begin{aligned} S_r &= C_r + \Lambda_r - B_r^\top K^{-1} B_r, \\ w_r &= q_r - B_r^\top K^{-1} \bar{f}, \\ \Delta_{\mathrm{McL}}(r;\tau) &= w_r^\top S_r^+ w_r \\ &= \big(q_r - B_r^\top K^{-1}\bar{f}\big)^\top \big(C_r + \Lambda_r - B_r^\top K^{-1} B_r\big)^+ \big(q_r - B_r^\top K^{-1}\bar{f}\big). \end{aligned}$$

So S_r is the metric of the new block after removing the part already explainable by the old tangent space, w_r is the residualized overlap with the exact projective drift, and $\Delta_{\mathrm{McL}}(r;\tau)$ is residualized force squared divided by residualized norm.

17B.11 The unregularized geometric meaning

Set $\Lambda = \Lambda_c = 0$. For a single real candidate direction $\bar{u}$, define

$$\bar{u}_\perp := (I - P_{\bar{T}}^{\mathbb{R}})\bar{u}, \qquad \bar{r}_0 := (I - P_{\bar{T}}^{\mathbb{R}})\bar{b}.$$

Then the exact one-direction gain closes to

$$\begin{aligned} \Delta_0 &= \frac{\big(\Re\langle \bar{u}_\perp, \bar{r}_0\rangle\big)^2}{\|\bar{u}_\perp\|^2} \\ &= \frac{\big(\Re\langle (I - P_{\bar{T}}^{\mathbb{R}})\bar{u}, (I - P_{\bar{T}}^{\mathbb{R}})\bar{b}\rangle\big)^2}{\|(I - P_{\bar{T}}^{\mathbb{R}})\bar{u}\|^2}. \end{aligned}$$

The numerator uses $\Re\langle \bar{u}_\perp, \bar{r}_0\rangle$ rather than $|\langle \bar{u}_\perp, \bar{r}_0\rangle|$ because one is adding **one real parameter**, not an arbitrary complex tangent line.

17B.12 When should the ansatz adapt?

The intrinsic normalized miss closes to

$$\begin{aligned} \rho_{\mathrm{miss}} &:= \frac{\epsilon_{\mathrm{proj}}^2}{\|\bar{b}\|^2 + \varepsilon} = \frac{\|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f}}{\|\bar{b}\|^2 + \varepsilon} \\ &= \frac{\mathrm{Var}_\psi(H) - \bar{f}^\top \bar{G}^+ \bar{f}}{\mathrm{Var}_\psi(H) + \varepsilon} = \frac{\langle \psi | (H - E_\psi)^2 | \psi\rangle - \bar{f}^\top \bar{G}^+ \bar{f}}{\langle \psi | (H - E_\psi)^2 | \psi\rangle + \varepsilon}. \end{aligned}$$

Interpretation:

- $\rho_{\mathrm{miss}} \approx 0$: the current manifold already captures the projective velocity well,
- ρ_{miss} large: the manifold is missing important directions.

A clean rule is therefore

$$\text{adapt if } \rho_{\mathrm{miss}} > \tau_{\mathrm{miss}}.$$

One may also adapt for numerical unreliability or after a maximum interval without any structural update.

17B.13 Which candidates should be admissible?

A candidate may have nonzero gain and still be numerically poor because it is nearly redundant. So before ranking by Δ_λ, one should inspect the residualized geometry:

- compute singular values of S, or of the corresponding whitened residualized block,
- reject candidates whose retained support is too ill-conditioned.

An invariant way to do this is to whiten the old tangent metric. If

$$\bar{G} \approx V\Sigma^2 V^\top$$

on its numerical support, define whitened coordinates

$$z := \Sigma V^\top \dot{\theta}.$$

Then penalizing $\|z\|^2$ is a penalty on **physical tangent speed** rather than coordinate speed. This is the least chart-dependent way to stabilize an overparameterized manifold.

17B.14 Two different residuals should be tracked

The projective residual is

$$\bar{r} := \bar{T}\dot{\theta}_\lambda - \bar{b}.$$

This is the correct gauge-invariant quantity for local candidate scoring and local manifold-growth decisions.

The raw Hilbert-space residual is

$$r_{\text{raw}} := T\dot{\theta}_\lambda - (-iH\psi) = T\dot{\theta}_\lambda + iH\psi.$$

These are related by

$$\bar{r} = Q_\psi r_{\text{raw}}.$$

So:

- use $\bar{r}$ for **local structure growth**,
- use r_{raw} when a rigorous Hilbert-space accumulated-error bound is desired.

17B.15 Why integrated residual is the right branch-pruning quantity

Let a branch be propagated on a fixed scaffold over a segment $[\tau_a, \tau_b]$. Let $|\psi_{\text{ex}}\rangle$ be the exact state and $|\psi_{\text{var}}\rangle$ the variational state with the same initial condition at τ_a, and define

$$e(\tau) := |\psi_{\text{var}}(\tau)\rangle - |\psi_{\text{ex}}(\tau)\rangle.$$

Then the error evolution, Duhamel representation, norm bound, and segment cost close to

$$\dot{\psi}_{\text{ex}} = -iH\psi_{\text{ex}}, \qquad \dot{\psi}_{\text{var}} = T\dot{\theta}_\lambda, \quad r_{\text{raw}} = T\dot{\theta}_\lambda + iH\psi,$$

$$\dot{e} = -iHe + r_{\text{raw}},$$

$$e(\tau) = \int_{\tau_a}^{\tau} U(\tau, s)\, r_{\text{raw}}(s)\, ds,$$

$$\|e(\tau)\| \le \int_{\tau_a}^{\tau} \|r_{\text{raw}}(s)\|\, ds \le \sqrt{\tau - \tau_a} \left(\int_{\tau_a}^{\tau} \|r_{\text{raw}}(s)\|^2\, ds \right)^{1/2},$$

$$J_{\text{seg}}^{\text{raw}} := \int_{\tau_a}^{\tau_b} \|r_{\text{raw}}(\tau)\|^2\, d\tau = \int_{\tau_a}^{\tau_b} \|T\dot{\theta}_\lambda(\tau) + iH(\tau)\psi(\tau)\|^2\, d\tau.$$

So $J_{\text{seg}}^{\text{raw}}$ is a natural upper-bound surrogate for accumulated state-vector error. If one wants a fully gauge-invariant pruning metric instead, replace $\|r_{\text{raw}}(\tau)\|^2$ by $\|\bar{r}(\tau)\|^2 = \|Q_\psi r_{\text{raw}}(\tau)\|^2$ and interpret the result as a ray/projective error surrogate.

17B.16 The branch objective

A branch $\mathfrak{b}$ is a sequence of scaffolds $\mathcal{O}_0, \mathcal{O}_1, \ldots, \mathcal{O}_J$ used over consecutive checkpoint segments, and the cumulative objective closes as

$$\begin{aligned}
J_{\text{dyn}}(\mathfrak{b}) &= \sum_{j=0}^{J-1} J_{\text{seg},j} + \beta\, C_{\text{hw}} + \gamma\, N_{\text{adapt}} + \mu_E \sum_{j=0}^{J-1} \int_{\tau_j}^{\tau_{j+1}} \left| \frac{d}{d\tau} \langle H(\tau)\rangle_j - \langle \partial_\tau H(\tau)\rangle_j \right|^2 d\tau \\
&= \sum_{j=0}^{J-1} \int_{\tau_j}^{\tau_{j+1}} \|r_{\text{raw}}^{(j)}(\tau)\|^2\, d\tau + \beta\, C_{\text{hw}} + \gamma\, N_{\text{adapt}} + \mu_E \sum_{j=0}^{J-1} \int_{\tau_j}^{\tau_{j+1}} \left| \frac{d}{d\tau} \langle H(\tau)\rangle_j - \langle \partial_\tau H(\tau)\rangle_j \right|^2 d\tau \\
&= \sum_{j=0}^{J-1} \int_{\tau_j}^{\tau_{j+1}} \left\| T^{(j)}(\tau)\dot{\theta}_\lambda^{(j)}(\tau) + iH(\tau)\psi^{(j)}(\tau) \right\|^2 d\tau + \beta\, C_{\text{hw}} + \gamma\, N_{\text{adapt}} + \mu_E \sum_{j=0}^{J-1} \int_{\tau_j}^{\tau_{j+1}} \left| \frac{d}{d\tau} \langle H(\tau)\rangle_j - \langle \partial_\tau H(\tau)\right.
\end{aligned}$$

If one wants a fully gauge-invariant branch surrogate instead, replace $\|r_{\text{raw}}^{(j)}(\tau)\|^2$ by $\|\bar{r}^{(j)}(\tau)\|^2 = \|Q_{\psi^{(j)}} r_{\text{raw}}^{(j)}(\tau)\|^2$.

17B.17 The full process, in one coherent sequence

At each physical time τ, with current scaffold $\mathcal{O}$, the full loop can be written compactly as

$$\begin{aligned}
&\text{A: } |\psi\rangle = |\psi(\theta;\mathcal{O})\rangle, \qquad Q_\psi = I - |\psi\rangle\langle\psi|, \qquad \bar{T} = Q_\psi T, \qquad \bar{b} = Q_\psi(-iH\psi) = -i(H - E_\psi)\psi, \\
&\text{B: } \bar{G} = \Re(\bar{T}^\dagger \bar{T}), \qquad \bar{f} = \Re(\bar{T}^\dagger \bar{b}), \qquad K = \bar{G} + \Lambda, \qquad \dot{\theta}_\lambda = K^+ \bar{f}, \\
&\quad \epsilon_{\text{proj}}^2 = \|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f}, \qquad \epsilon_{\text{step}}^2 = \|\bar{T}\dot{\theta}_\lambda - \bar{b}\|^2, \qquad \bar{r} = \bar{T}\dot{\theta}_\lambda - \bar{b}, \qquad r_{\text{raw}} = T\dot{\theta}_\lambda + iH\psi, \\
&\text{C: } \rho_{\text{miss}} = \frac{\epsilon_{\text{proj}}^2}{\|\bar{b}\|^2 + \varepsilon} = \frac{\epsilon_{\text{proj}}^2}{\text{Var}_\psi(H) + \varepsilon}, \qquad \text{adapt iff } \rho_{\text{miss}} > \tau_{\text{miss}}, \\
&\text{D: } B_r = \Re(\bar{T}^\dagger \bar{U}_r), \qquad C_r = \Re(\bar{U}_r^\dagger \bar{U}_r), \qquad q_r = \Re(\bar{U}_r^\dagger \bar{b}), \\
&\quad S_r = C_r + \Lambda_r - B_r^\top K^+ B_r, \qquad w_r = q_r - B_r^\top K^+ \bar{f}, \qquad \Delta_{\text{McL}}(r;\tau) = w_r^\top S_r^+ w_r.
\end{aligned}$$

For each shortlisted admissible candidate-position record $r = (m, p)$, one then creates a zero-initialized child, propagates it over the next probe segment, and prunes on the accumulated branch objective:

$$\begin{aligned}
&\text{E: } \mathcal{O} \mapsto \mathcal{O}_r^+, \qquad |\psi(\theta, 0; \mathcal{O}_r^+)\rangle = |\psi(\theta;\mathcal{O})\rangle, \\
&\text{F: } J_{\text{seg},r}^{\text{raw}} = \int_{\tau_a}^{\tau_b} \|r_{\text{raw},r}(\tau)\|^2\, d\tau = \int_{\tau_a}^{\tau_b} \left\| T_r(\tau)\dot{\theta}_{\lambda,r}(\tau) + iH(\tau)\psi_r(\tau) \right\|^2 d\tau, \\
&\text{G: } J_{\text{dyn}}^{\text{new}}(r;\mathfrak{b}) = J_{\text{dyn}}^{\text{old}}(\mathfrak{b}) + J_{\text{seg},r}^{\text{raw}} + \beta\, C_{\text{hw},r} + \gamma\, N_{\text{adapt},r}, \qquad \mathfrak{B}_{\text{next}} = \text{Prune}\Big(\{(\mathcal{O}_r^+, J_{\text{dyn}}^{\text{new}}(r;\mathfrak{b}))\}_r\Big).
\end{aligned}$$

That is the whole piecewise-manifold control loop.

17B.18 One concise interpretation

The method is doing this:

The exact quantum dynamics asks for a projective velocity $\bar{b}$.
The current ansatz only provides a real tangent plane.
McLachlan chooses the closest available tangent velocity.
If the miss is too large, the algorithm enlarges the tangent plane by adding zero-initialized candidate directions.
The local gain formula tells us which candidate most reduces the current miss.
Short probe rollouts then decide which structural choices remain best after actual propagation.

17B.19 Explicit symbolic–mathematical differences between the clean story and the current implementation

The clean story above is the right mathematical idealization. The current implementation differs from that idealization in a few explicit ways:

1. **Sign convention.** The clean story uses

$$\bar{b} = -iQ_\psi H\psi, \qquad \bar{r} = \bar{T}\dot{\theta} - \bar{b}.$$

The implementation instead uses

$$r = \bar{T}\dot{\theta} + iQ_\psi H\psi, \qquad f_i = \Im\langle \bar{\tau}_i | H | \psi\rangle, \qquad \Re\langle \bar{\tau}_i, -iH\psi\rangle = \Im\langle \bar{\tau}_i, H\psi\rangle.$$

1. **Regularization surface.** Rather than a general positive-semidefinite penalty Λ or Λ_c, the implementation uses a scalar $\lambda \geq 0$ on either chart coordinates or whitened metric coordinates:

$$\begin{aligned} K_{\text{chart}} &= \bar{G} + \lambda I, \\ \bar{G} &\approx V\Sigma^2 V^\top, \qquad z = \Sigma V^\top \dot{\theta}, \qquad (I + \lambda I)z = \Sigma^{-1} V^\top \bar{f}, \qquad \dot{\theta} = V\Sigma^{-1} z. \end{aligned}$$

1. **Propagation solve versus policy solve.** The clean story presents one stabilized local solve, whereas the implementation may separate the propagated step from the policy/telemetry step:

$$(\dot{\theta}_{\text{prop}}, \epsilon^{2,\text{prop}}_{\text{step}}) \neq (\dot{\theta}_{\text{policy}}, \epsilon^{2,\text{policy}}_{\text{step}}) \quad \text{in general,}$$

even when both are derived from the same local geometry.

1. **Variance normalization and guards.** The clean story writes

$$\rho_{\text{miss}} = \frac{\epsilon^2_{\text{proj}}}{\text{Var}_\psi(H) + \varepsilon}.$$

The implemented ratio surface is closer to

$$\rho_{\text{miss}} = \begin{cases} \epsilon^2_{\text{proj}}/(\text{Var}_\psi(H) + \varepsilon), & \text{Var}_\psi(H) \geq \nu_{\min}, \\ \text{suppressed}, & \text{Var}_\psi(H) < \nu_{\min}, \end{cases}$$

with an auxiliary guard flag recording the suppressed regime.

1. **Active-parameter windows.** The clean story is written on the full tangent space, but the implementation may restrict to an active suffix or oracle-supplied index set:

$$T \mapsto T_{W_{\mathrm{act}}}, \qquad \dot{\theta} \mapsto (\dot{\theta}_{W_{\mathrm{act}}}, 0), \qquad \bar{G} \mapsto \bar{G}_{W_{\mathrm{act}} W_{\mathrm{act}}},$$

so inactive coordinates are frozen.

1. **Candidate gain evaluation.** The clean story scores candidates by the Schur-complement gain

$$\Delta_\lambda = w^\top S^+ w.$$

The implementation instead zero-initializes one append candidate, rebuilds the augmented local geometry, and scores by direct defect drops

$$\Delta_0 = \epsilon^2_{\mathrm{proj,current}} - \epsilon^2_{\mathrm{proj,aug}}, \qquad \Delta_\lambda = \epsilon^2_{\mathrm{step,current}} - \epsilon^2_{\mathrm{step,aug}}.$$

1. **Position-aware current branch growth.** The clean story allows arbitrary insertion positions and multi-parameter blocks. The helper layer now supports position-aware insertion and position-centered post-insert refit windows,

$$\mathcal{T}(\mathfrak{b}, m, p) = \mathfrak{b}', \qquad \mathcal{O}_{b'} = \mathcal{O}_b \oplus_p m, \qquad \theta_{b'} = \theta_b \oplus_p 0,$$

and the active probe set is built from the append slot, the left boundary, and the active reoptimization window. Hence, when the active window is made maximally large and the probe-position cap is also made large enough, the implementation can evaluate the full insertion set $p \in \{0, \ldots, |\theta|\}$ rather than append only. The live surface still remains zero-initialized **single-parameter** growth, but not append-only growth.

1. **Checkpoint-trigger logic.** The clean story says "adapt when ρ_{miss} is too large." The implemented trigger surface is richer:

$$\mathbf{1}_{\mathrm{checkpoint}} = \mathbf{1}_{\mathrm{cadence}} \vee \mathbf{1}(\epsilon^2_{\mathrm{proj}} > \tau_{\mathrm{proj}}) \vee \mathbf{1}(\epsilon^2_{\mathrm{step}} > \tau_{\mathrm{step}}) \vee \mathbf{1}(\rho_{\mathrm{miss}} > \tau_{\mathrm{miss}} \text{ when defined}),$$

together with a model-versus-damping failure classification.

1. **Branch objective surface.** The clean story uses a scalar objective such as

$$J_{\mathrm{scalar}} = \sum J_{\mathrm{seg}} + \beta C_{\mathrm{hw}} + \gamma N_{\mathrm{adapt}}.$$

The implementation instead prunes by the lexicographic key

$$\kappa_{\mathrm{prune}} = \big(J^{\mathrm{rt}}, J^{\mathrm{res}}, J^{\mathrm{model}}, J^{\mathrm{damp}}, J^{\mathrm{cons}}, |\mathcal{O}|, k, \mathrm{labels}(\mathcal{O}), \mathrm{round}_{10}(\theta), \mathrm{id}\big),$$

after deduplicating by time index, selected-operator labels, and rounded parameter values.

1. **Projective versus raw residual accumulation.** The clean story distinguishes a projective residual $\bar{r}$ from a raw Hilbert-space residual r_{raw}. The implemented rollout telemetry is closer to

$$(\Delta J^{\epsilon}, \Delta J^{\text{model}}, \Delta J^{\text{damp}}) = \left(\int \epsilon^2_{\text{step}}, \int \epsilon^2_{\text{proj}}, \int \max\!\big(\epsilon^2_{\text{step}} - \epsilon^2_{\text{proj}}, 0\big) \right),$$

rather than a separately stored raw Hilbert-space residual bound.

1. **Geometry source.** The clean story is written as if the tangent vectors and $H\psi$ are assembled directly every time. The implementation may instead start from externally supplied local data:

$$(\bar{G}, \bar{f}, \mathrm{Var}_{\psi}(H)) = \mathcal{O}_{\text{geom}}(\psi, H, \mathcal{O}),$$

after which the host solves the McLachlan system on that returned geometry.

So the mathematical backbone is the same, but the implemented restricted surface closes schematically to

projective policy geometry + scalar chart/whitened damping + append-only zero-init growth
+ checkpointed short-horizon rollout + lexicographic prune key + optional oracle-sup

17B.20 Discrete spectra and density-sector observables for HH checkpoint dynamics

A controller trajectory row r_n (index $n = 0, \ldots, N-1$) carries either `time` or `physical_time`, and stores raw lattice observables under keys such as `site_occupations`, `site_occupations_exact`, `staggered`, `doublon`, and `doublon_exact`. The spectral stack in the repository works directly on these rows and does not re-simulate dynamics.

$$t_n := \mathrm{row}_n[\mathrm{time}_\star], \qquad \mathrm{time}_\star \in \{\texttt{time}, \texttt{physical_time}\}.$$

For each site $j \in \{0, \ldots, L-1\}$, define spin-resolved occupations

$$n_j^{\uparrow}(t_n) = \langle \psi(t_n) | \hat{n}_{j,\uparrow} | \psi(t_n) \rangle, \quad n_j^{\downarrow}(t_n) = \langle \psi(t_n) | \hat{n}_{j,\downarrow} | \psi(t_n) \rangle, \quad n_j(t_n) = n_j^{\uparrow}(t_n) + n_j^{\downarrow}(t_n),$$

with $\hat{n}_{j,\sigma} = (I - \hat{Z}_{j,\sigma})/2$ under this codebase's convention. Equivalently, in the statevector path used by `hh_realtime_checkpoint_controller.py`, these are evaluated as

$$n_j^{\sigma}(t_n) = \sum_s p_s \, b_j^{\sigma}(s), \qquad D(t_n) = \sum_s p_s \sum_{j=0}^{L-1} b_j^{\uparrow}(s) b_j^{\downarrow}(s),$$

where $p_s = |\psi_s|^2$ in the computational basis and $b_j^{\sigma}(s) \in \{0, 1\}$ are basis bits. The repository therefore records

$$\texttt{site_occupations}[n, j] = n_j(t_n), \qquad \texttt{staggered}[n] = \frac{1}{L} \sum_{j=0}^{L-1} (-1)^j n_j(t_n), \qquad \texttt{doublon}[n] = D(t_n),$$

and the corresponding `_exact` fields for the driven exact trajectory.

$$D(t_n) = \sum_{j=0}^{L-1} n_j^{\uparrow}(t_n) n_j^{\downarrow}(t_n).$$

Here j denotes the lattice-site index, while n denotes the discrete time-sample index.

From $\{n_j(t_n)\}$ one obtains the secondary diagnostics used in the spectrum module:

$$\delta n_j(t_n) := n_j(t_n) - \frac{1}{L}\sum_{m=0}^{L-1} n_m(t_n), \qquad d_{ij}(t_n) := n_i(t_n) - n_j(t_n), \qquad m(t_n) := \frac{1}{L}\sum_{j=0}^{L-1}(-1)^j n_j(t_n).$$

(For $L = 2$, $\delta n_0 = \frac{1}{2}\, d_{01}$ and $\delta n_1 = -\frac{1}{2}\, d_{01}$.)

For spectral post-processing, let $x_n := x(t_n)$ denote any sampled scalar observable. The implemented V1 pipeline first infers or assumes a uniform step $t_{n+1} - t_n = \Delta t$, then detrends and re-centers the trace, applies either no window or a Hann window, and finally evaluates the one-sided real FFT. In this notation, $\tilde{x}_n$ denotes the detrended and re-centered signal: for constant detrending it is x_n with its sample mean removed, whereas for linear detrending it is the residual after subtracting the least-squares line in t_n, re-centered exactly as in the code path. The windowed sequence is $y_n := w_n \tilde{x}_n$, with $w_n = 1$ for no window and w_n equal to the standard Hann taper for Hann windowing.

$$Y_k = \sum_{n=0}^{N-1} y_n e^{-i2\pi kn/N}, \qquad k = 0, \ldots, \lfloor N/2 \rfloor,$$

which is the one-sided coefficient returned by the real-FFT stage (implemented with `np.fft.rfft`) for a real sampled trace. The associated frequency grid is

$$\omega_k = \frac{2\pi k}{N\Delta t}, \qquad f_k = \frac{k}{N\Delta t}, \qquad \omega_{\mathrm{Ny}} = \frac{\pi}{\Delta t}, \qquad \Delta\omega = \frac{2\pi}{N\Delta t}.$$

The reported one-sided amplitude A_k is obtained from $|Y_k|$ by normalizing with the window-weight sum and doubling positive-frequency bins away from DC and Nyquist:

$$A_k = \begin{cases} \dfrac{|Y_k|}{\sum_{n=0}^{N-1} w_n}, & k = 0, \\ \dfrac{2|Y_k|}{\sum_{n=0}^{N-1} w_n}, & 1 \le k < N/2, \\ \dfrac{|Y_k|}{\sum_{n=0}^{N-1} w_n}, & k = N/2 \text{ when } N \text{ is even.} \end{cases}$$

Error-time signals are also treated as observables:

$$x_n^{\mathrm{err}} = x_n^{\mathrm{ctrl}} - x_n^{\mathrm{exact}} \quad \Rightarrow \quad \text{apply the same FFT-processing machinery to } x_n^{\mathrm{err}}.$$

When the drive frequency ω_d is known, the same centered trace is also fit to drive-locked harmonics:

$$x(t) \approx c_0 + \sum_{\ell=1}^{K_{\max}} \big(a_\ell \cos(\ell\omega_d t) + b_\ell \sin(\ell\omega_d t)\big), \quad R_\ell = \sqrt{a_\ell^2 + b_\ell^2},\ \phi_\ell = \mathrm{atan2}(b_\ell, a_\ell).$$

Here R_ℓ is the amplitude of the ℓ-th drive-locked harmonic and ϕ_ℓ is its phase.

The spectral stack is therefore a full transform layer on top of trajectory observables: raw site-resolved occupations, staggered density, doublon, and pair imbalance are each converted to sampled series, then detrend-window-FFT processed in the same one-sided-amplitude form, with controller-versus-exact residuals using the same operators.